Over the Hills and Far Away: Macroprudential Policy and Firm Foreign Exchange Exposure

Drishan Banerjee*

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Abstract

Macroprudential policy that shores up the domestic financial system may leave firms more exposed to external risk. I document this trade-off using balance sheet data for Indian non-financial firms. When macroprudential measures tighten the supply of domestic credit, bank-dependent firms face sharp credit rationing and substitute toward external borrowing in foreign currency. This exposes firms to exchange rate fluctuations. Firms that hold more foreign currency debt suffer larger losses, lower profits and a higher likelihood of distress when the rupee depreciates. I build a two-sector model in which firms can replace bank loans with foreign currency debt, and use it to analyze the implications. The model captures the trade-off between greater resilience of domestic banks and greater fragility of firms due to currency mismatch on their balance sheets, and is used to quantify the welfare consequences of macroprudential regulations and derive their optimal stringency. Calibrated to India, the welfare-maximizing capital requirement is close to the Reserve Bank of India's current minimum. The results highlight the importance of considering cross-market and currency substitution when designing macroprudential policy in emerging economies.

Keywords: Macroprudential Policy, Foreign Currency Debt, Bank Capital Requirements

JEL Codes: E44, F34, G28, G32

*UC Santa Cruz. drbanerj@ucsc.edu. I am grateful to Galina Hale, Alonso Villacorta, Kenneth Kletzer, Rajeswari Sengupta and Pascal Michaillat for valuable suggestions. All errors are mine.

1 Introduction

Macroprudential policy aims to reduce systemic risk by strengthening the financial system. The Global Financial Crisis led to a wider use of these policies and renewed interest in their effects (Brunnermeier et al. 2009, Hanson et al. 2011). These tools are now widely regarded as essential instruments for central banks and financial regulators seeking to contain systemic risk.[1] Yet, policies that strengthen banks may not leave the firms that borrow from it equally protected. When regulation restricts domestic bank credit, firms may turn to foreign lenders and borrow in foreign currency. This shifts risk from banks to firms and exposes their balance sheets to exchange rate shocks. This possibility has so far received limited direct attention and has not been closely analyzed in the academic or policy literature. This paper examines whether such substitution takes place, and what it implies for the appropriate stance of macroprudential policy.

Most macroprudential instruments, especially in emerging economies, operate through lenders rather than borrowers. These instruments change the ability of domestic banks to supply credit without directly regulating firms (Cerutti et al. 2017, Yang and Suh 2023). By restricting bank lending, these measures can limit credit growth, reduce the cost of a downturn and reinforce the overall stability of the financial system. But firms may respond by borrowing abroad. Since foreign borrowing is often denominated in foreign currency, a depreciation of the domestic currency raises debt-servicing costs and increases the risk of default.[2] If these defaults cause losses for domestic banks, the policy may create a source of systemic risk even as it strengthens bank balance sheets (Niepmann and Schmidt-Eisenlohr 2022, Kim et al. 2015, Bruno and Shin 2020, Alfaro et al. 2019). Thus, the stability gained by the banking system is in part traded for fragility on the corporate side.

Against this background, the paper asks two questions. First, does macroprudential tightening that restricts bank credit lead firms that depend on banks to substitute toward foreign-currency debt, and does this exposure increase their vulnerability to

[1] Mendoza (2016). See Galati and Moessner (2013) for a comprehensive survey.

[2] Many emerging economies find it difficult to borrow abroad in their own currency, a pattern Eichengreen et al. (2007) call original sin. Many countries with sound fundamentals still suffer from it and it is governed mainly by country size and the structure of global portfolios where international borrowing is concentrated in a few major currencies (Eichengreen et al. 2007, Hausmann and Panizza 2003). Several middle-income countries have since built local-currency government bond markets but original sin persists and the currency mismatch has shifted within these economies rather than disappeared (Eichengreen et al. 2023). For firms, foreign borrowing remains mostly in foreign currency and greater reliance on it by the corporate sector raises country risk (Du and Schreger 2022).

currency depreciation? Second, how should a regulator set capital requirements to balance the gains from a more resilient financial system against the costs of greater firm fragility?

India provides a useful setting to study this trade-off. It is one of the most active users of macroprudential policy, second only to China in the number of actions taken, and almost all its measures are lender-based. Since banks are the main source of credit for firms, regulations that constrain bank lending can affect firms' access to finance. Indian firms can turn to foreign lenders, but this borrowing is largely denominated in foreign currency. About two-thirds of the firms in my sample with foreign-currency debt owe more in foreign currency than they earn from exports in a year. They therefore have limited natural hedges against exchange rate risk. Hedging through currency derivatives is also costly and constrained by market liquidity (Patnaik et al. 2015, Goel et al. 2025).

I proceed in three steps. First, I use firm-level data from India to test whether firms that depend more on domestic banks shift toward foreign-currency borrowing when macroprudential policy tightens. Second, I examine whether foreign-currency exposure increases firm losses and distress when the rupee depreciates. Third, I build a model of firms and banks that incorporates this substitution channel and use it to determine the welfare-maximising capital requirement. I then calibrate the model to India and compare the implied requirement with the one in place.

For the empirical analysis, I use firm-level balance sheet data for Indian non-financial firms from the Center for Monitoring the Indian Economy's (CMIE) Prowess database together with the macroprudential policy stance measured by the Macroprudential Policy Index (Alam et al. 2019). A firm's exposure to bank-specific macroprudential policy depends on how much it relies on bank credit to begin with. Hence, I exploit differences in firms' initial dependence on bank credit to identify their responses to policy tightening.

The empirical analysis shows that with macroprudential tightening, firms that rely more on domestic banks reduce the share of bank borrowing and increase the share of foreign-currency debt. Their total borrowing remains largely unchanged. The results suggest that firms replace the credit they lose from domestic banks rather than reduce their borrowing. This extends the literature on credit substitution, which has looked at firms substituting across sources of credit rather than scaling back when bank lending contracts (Adrian et al. 2013, Becker and Ivashina 2014, Crouzet 2018). However, this literature has focused on substitution within domestic markets, from bank loans to

corporate bonds in advanced economies and says little about substitution into foreign-currency debt. In emerging economies, the shift toward foreign-currency debt matters because it changes the currency composition of firms' liabilities and exposes them to exchange rate risk.

I show that firms with more foreign-currency debt relative to assets incur higher foreign exchange losses when the Indian rupee depreciates and also experience lower profitability and return on assets. They are also more likely to have operating earnings below their interest expenses. For a firm with 10 percentage points more foreign-currency debt relative to assets, a 10 percent depreciation raises foreign exchange losses by about 0.11 percentage points of assets. It also reduces operating profitability and return on assets while raising the probability of distress by about 7 per cent. Taken together, these results link macroprudential tightening to firm fragility through two channels: tightening raises firms' foreign currency exposure, and that exposure turns into losses and distress when the currency depreciates.

Having established this stylized fact, I analyze its consequences for financial stability and for optimal capital requirement. I build a two-period small open economy model of firms and banks. Firms borrow from domestic banks and foreign lenders, while banks lend to firms and finance themselves through deposits. A higher capital requirement raises the cost of domestic credit and induces firms to substitute toward foreign-currency debt. This makes firms more vulnerable when the domestic currency depreciates. If these firms default, domestic banks incur losses on the loans that remain on their balance sheets. The risk created by foreign borrowing therefore feeds back into the banking system that macroprudential policy seeks to protect. This feedback is particularly relevant for emerging market economies, where firms often borrow abroad in foreign currency.[3]. At the same time, higher capital requirements leave banks well capitalized and better equipped to withstand a downturn and continue lending at a cheaper price when firms need credit to recover. Hence the downturn is less severe. The benefit of the macroprudential policy is best described as a reallocation of credit across states rather than an increase in it. The policy thus has two opposing effects: it increases firms' exposure to exchange rate risk in normal times by withholding domestic intermediation capacity and it returns the capacity in a crisis when the net worth of

[3] Although Latin American and Asian emerging markets launched initiatives to develop local currency sovereign bond markets to address the currency and maturity mismatches of the 1990s, firms however, still borrow abroad largely in foreign currency. Non-financial corporations in emerging Asia and Latin America sharply increased their international debt issuance after 2009, most of it in US dollars and often through overseas subsidiaries (Chui et al. 2014, Bruno and Shin 2017, Du and Schreger 2022)

both banks and firms have been eroded and firms need credit to recover. The optimal requirement balances these costs and benefits.

The model also captures why banks may hold less capital than is socially optimal. An individual bank has an incentive to lend more in normal times rather than hold capital that could support lending during a future crisis. When a crisis hits, capital is scarce and the banks that still have it can lend at high rates, so no single bank gains from the lower borrowing costs that higher capital across the entire banking system would bring to firms. It does not fully account for the benefits that higher capital across the entire banking system creates for firms when a crisis occurs. A regulator can address this externality by requiring banks to hold capital that they would not choose to hold on their own. But a higher requirement also raises the cost of domestic credit and pushes firms toward foreign-currency borrowing. For the optimal stance of macroprudential policy, the regulator must therefore weigh the gains from credit supplied during a crisis against the loss of credit and the increase in firm fragility during normal times.

The calibration to India shows that the welfare-maximising capital requirement is close to the Reserve Bank of India's minimum. The model reproduces the substitution toward foreign-currency debt that is found in the data and also the increase in firm distress when the rupee depreciates. As the capital requirement rises, banks become better able to supply credit after a crisis, but firms become more exposed to exchange rate risk. A requirement above about 3 percent is enough to keep banks solvent in every state, while banks left to their own decisions hold capital of about 5.6 percent. However, preventing bank failure does not by itself maximise welfare. The gains from higher capital continue because banks can supply credit when firms need it most. They stop when these gains are offset by the cost of reduced domestic credit and the fragility created by firms' foreign-currency borrowing. The welfare-maximising requirement is about 8.4 percent, close to the RBI's minimum requirement which is 9 percent.

This paper makes three contributions. First, it brings firm-level evidence to the study of leakage from macroprudential policy. Existing studies show that tighter regulation of domestic banks can shift credit toward foreign lenders or non-bank financial institutions. [4] These studies focus on lenders or aggregate credit and do not trace how firms respond to the change in credit supply. Additionally, leakage is usually seen as

[4] Foreign bank branches in the United Kingdom made up about a third of the drop in lending that followed higher capital requirements on domestic banks (Aiyar et al. 2014). Borrowing from foreign banks rises after capital and reserve requirements tighten (Reinhardt and Sowerbutts 2015, Avdjiev et al. 2016, Cerutti et al. 2017), and credit moves toward non bank lenders (Cizel et al. 2019).

a concern when the aim is to slow credit growth, but less so when the aim is a safer banking system, since banks hold more capital either way (Reinhardt and Sowerbutts 2015). I contribute to this literature by showing that the source of replacement credit matters even for the second aim: when firms substitute toward foreign-currency borrowing, the resulting exchange rate risk can feed back into domestic banks through firm defaults.

Second, I show that bank regulation with no currency dimension can still create foreign currency exposure. The closest evidence studies regulations aimed at currency exposure itself. Ahnert et al. (2021) study regulations that directly target bank's foreign currency lending and funding. In contrast, I show that restrictions on domestic bank credit can increase firms' foreign-currency exposure by inducing them to borrow abroad.[5] This identifies domestic regulation as a source of foreign-currency borrowing, which is usually traced to forces outside a country's control.[6] In India, Acharya and Vij (2020) show that firms borrow more abroad when the dollar carry trade is profitable, a pull from global markets. This paper complements their study by showing that domestic macroprudential regulations can act as a push factor.

Third, I incorporate this substitution channel into a model of optimal capital requirements. Existing models such as Elenev et al. (2021) weigh the benefits of bank resilience against the cost of reduced credit, but they study this in a closed economy. Falasconi et al. (2024) study capital requirements in a small open economy, but there the banks borrow abroad: safer banks get cheaper foreign funding and take on more of it. In contrast, in my model, firms, not banks, borrow in foreign currency. When higher capital requirements raise the cost of domestic credit, firms turn to foreign lenders. If the rupee depreciates, these firms may default, transmitting losses to domestic banks. Tighter capital requirements therefore create a cost that closed-economy models overlook. The regulator must weigh this cost against the benefit of better-capitalised banks being able to sustain credit during a crisis.

Section 2 describes the data and the institutional framework, Section 3 presents the empirical specification and stylized facts, Sections 4 to 6 presents the model and its results, Section 7 concludes.

[5] In advanced economies, firms replace bank loans with bonds instead. Crouzet (2018) shows that this leaves them more exposed to distress, and leaves open the question, whether changes in bank regulation would have the same effect.

[6] These include original sin, global liquidity and the carry trade (Eichengreen et al. 2003, Bruno and Shin 2015).

2 Data and Institutional Setting

This section describes the data and the setting in which macroprudential policy operates in India. The analysis combines three sources. Macroprudential policy actions are from the integrated Macroprudential Policy Database (iMaPP) compiled by the IMF. Firm balance sheet data from Centre for Monitoring the Indian Economy (CMIE) Prowess database. The rupee-dollar exchange rate and India's policy rate comes from FRED. I first describe India's macroprudential framework and how the Reserve Bank of India has used it over the sample period. I then explain how I build the measure of the policy stance from the iMaPP records. Finally, I describe the firm sample, the borrowing and control variables for the empirical analysis.

2.1 India's Macroprudential Framework

India provides a well-suited environment to study the effect of macroprudential policy on firm borrowing. It is a bank-dominated economy, so when the Reserve Bank of India (RBI) tightens regulations on banks it tightens the main source of credit for most firms. The RBI has used macroprudential instruments actively and continuously since the early 2000s and its framework is overwhelmingly lender based. The principal tools used by the central bank are capital buffers and capital requirements, reserve and liquidity requirements and sectoral risk weights. These measures operate on the balance sheets of lenders (in this case, banks), affecting the supply of loanable credit rather than on borrowers directly. Of the 102 actions taken by India since 2003, 95 are lender oriented. This design is central to the identification strategy because the instruments act on bank lending capacity and not on firms. The policy stance can be treated as exogenous to any individual borrower and its effect on firms operates through the supply of domestic credit. Figure 1 plots the number of these actions in each year, separated into tightening and easing. India is among the most active users of macroprudential policy in the world, being second only to China. While China's actions run heavily in one direction, India's are far more evenly divided between tightening and easing episodes providing genuine variation in the policy stance. India has notably had two distinct credit cycles: a period of expansion through the late 2000s against which the RBI leaned by tightening and a gradual slowdown which was characterized by a looser macroprudential stance. Like most emerging markets, the overall cumulative stance across the years has been one of tightening.

2.2 Macroprudential Policy Data

I measure the policy stance using the integrated Macroprudential Policy (iMaPP) database of Alam et al. (2019), which records the timing and direction of macroprudential actions across all sets of policy instruments. I construct a cumulative index as the running sum of net actions coded +1 to indicate a tightening and -1 to indicate a loosening. Since Limits on FX positions target firms' foreign currency exposure directly and would move mechanically with the outcome of Index, I exclude this in the index. Additionally, I also exclude borrower based intruments such as Loan to Value limits (LTV), Debt Service to Income (DSTI) and Debt to Income (DTI) requirements. The impulse used in the empirical analysis is the net annual change in the index over each firm's fiscal year. Figure 1 shows the evolution of the index over time for India from 2003 to 2022.

2.3 Firm Data

The firm level data is drawn from the Centre for Monitoring the Indian Economy's (CMIE) Prowess database, which has balance sheet data for listed Indian non-financial firms between 2003 and 2022. Firm counts are sparse in the 1990s and starting in 2003 keeps the analysis with rich firm counts and within the period of active macroprudential tightening. [7]

From the balance sheet, I focus on four borrowing variables: bank borrowings, foreign currency borrowings, borrowings from other financial institutions and total borrowings. The outcomes of interest are the composition of firm debt – the share of borrowing denominated in foreign currency, from banks and from other non-bank financial institutions. [8] I also take standard firm level controls commonly used in the corporate finance literature as firm size, measured as the logarithm of total assets; leverage, total borrowings as a share of assets; profitability, profit after tax relative to assets; and asset tangibility, net fixed assets relative to assets.

A firm's exposure to lender based policy depends on how much it relies on bank credit to begin with. I capture this with a predetermined measure of bank dependence, fixed at the initial level of a firm's bank debt to assets. This reflects the firm's initial reliance on domestic credit from banks and is held constant so that it is unaffected

[7] Observations are aligned to each firm's fiscal-year-end rather than the calendar year. The panel is cleaned to one observation per firm-year, retaining firms with positive reported assets.

[8] Debt ratios are expressed in percentage points and winsorized at the 1st and 99th percentiles.

by any later policy induced adjustment that can introduce endogeneity. Because the ability to substitute toward foreign currency requires access to foreign credit markets in the first place, the analysis focuses on firms that borrow in foreign currency at some point in the sample, for which the substitution margin is actually available.[9]

Table 1 reports the summary statistics for the firms in the sample. The firms who have access to foreign currency debt are the large, established firms. They comprise the core of the Indian corporate sector rather than marginal fringe firms. The median firm who has access to foreign debt markets holds several times over the assets of the median firm without foreign access. They are significantly more profitable with a much larger fixed capital base. At the same time, these firms also lean heavily on domestic banks. Bank loans make up the largest part of their borrowing. This is what makes the substitution margin relevant, since a firm with a large bank-debt book to refinance and the market access to borrow abroad has both the motive and the means to move once bank credit is rationed. Table 2 looks more closely at the foreign currency borrowing of these firms. A sizeable part of the balance sheet, a median of close to 30% of total borrowing and 9% of assets is in the form of foreign currency debt. This borrowing is largely a financing choice rather than a hedge since close to half of the firms earn no export revenue at all. And among those that hold such debt, the median export share of sales is only 2%. Putting the two sides of the balance sheet together, about two-thirds of foreign-currency borrowers owe more in foreign currency than they earn in exports over a year. Their foreign-currency debt therefore has no natural revenue to offset it, so a depreciation raises the burden of the debt without a matching rise in income. This is the currency mismatch that turns the substitution this paper documents into a genuine source of risk.

[9] This sample retains firms with demonstrated access to foreign-currency funding while excluding those whose absence of FX borrowing is mechanical. Within it, roughly thirty percent of firms carry positive foreign-currency debt primarily in the form of external commercial borrowings, providing ample within-firm variation.

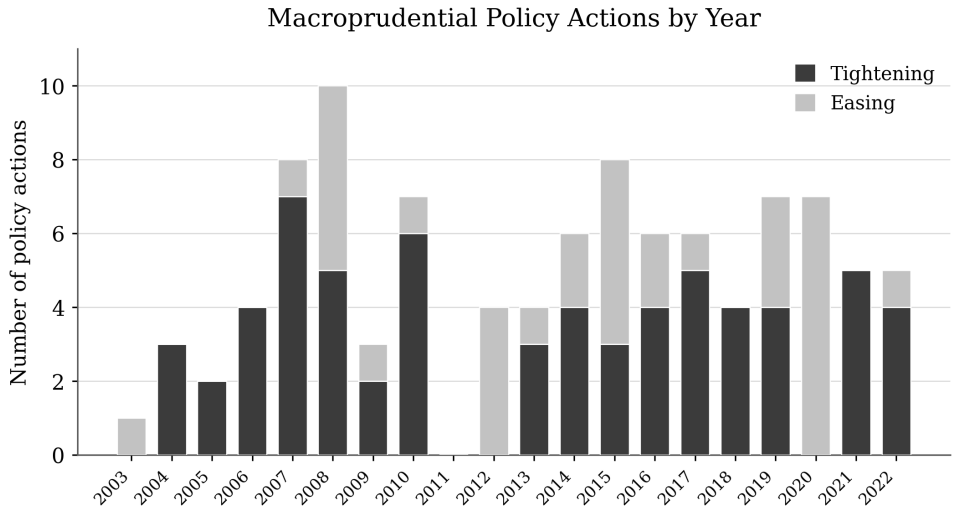


Figure 1: Macroprudential Policy Actions by Year

Notes: Number of macroprudential policy actions taken by India in each year between 2003 and 2022, from the iMaPP database (Alam et al. 2019). Each bar is separated into tightening (dark) and easing (light) actions, summed across the instruments recorded in the database. The figure counts all recorded actions; the policy index used in the estimation excludes limits on foreign-exchange positions. Source: iMaPP database.

3 Empirics

This section tests whether macroprudential tightening pushes bank dependent firms towards foreign currency debt. I first set out the local projection specification and show how the composition of firm borrowing responds to a tightening and what this means for firms' borrowing costs, followed by a set of robustness checks. Finally, I ask whether foreign currency exposure is costly by looking at how exposed firms fare when the rupee depreciates.

3.1 Empirical Specification

I estimate the response of firm borrowing to a macroprudential tightening using the local projection method of Jordà (2023). A firm's exposure to lender-based macroprudential policy is not random. It depends on how much the firm relies on domestic bank credit to begin with. To exploit this heterogeneity within firms, I let the response of a tightening to differ with each firm's initial dependence on banks. For each horizon $h = 0, ..., 5$ I estimate

Table 1: Descriptive statistics: all firms, and firms with and without access to foreign credit

	All Firms			Firms with Access to Foreign Credit			Firms without Access to Foreign Credit		
Variable	Mean	SD	Med.	Mean	SD	Med.	Mean	SD	Med.
Total assets (Rs. mn)	3,547	13,330	252	10,190	22,434	1,881	2,495	10,865	179
Capital employed (Rs. mn)	2,017	7,069	146	6,249	12,649	1,240	1,342	5,396	102
Net worth (Rs. mn)	1,084	4,131	66	3,130	6,988	531	758	3,351	48
Tangible net worth (Rs. mn)	1,041	4,011	63	3,022	6,793	516	726	3,251	47
Net fixed assets (Rs. mn)	838	3,161	44	2,692	5,735	486	484	2,202	27
Total borrowings (Rs. mn)	1,334	4,573	115	3,452	7,692	562	833	3,250	77
Sales (Rs. mn)	2,664	8,020	304	6,918	13,249	1,721	1,706	5,855	188
Log assets	5.40	2.64	5.53	7.54	2.05	7.54	5.06	2.57	5.19
Leverage, borrowings/assets (%)	59.9	122.2	35.5	47.2	82.9	34.9	62.9	129.6	35.8
Bank-debt share of borrowings (%)	66.4	31.9	75.0	47.6	27.4	48.6	67.4	32.1	76.9
FI/NBFC share of borrowings (%)	27.2	29.7	15.2	14.7	17.8	7.8	29.6	31.0	17.4
FX debt / assets (%)	0.5	2.8	0.0	3.6	6.8	0.0	0.0	0.0	0.0
Profitability, PBDITA/assets (%)	7.3	16.2	6.9	10.4	13.0	10.6	6.8	16.7	6.2
Tangibility, NFA/assets (%)	25.7	25.9	17.5	32.8	22.3	30.0	24.4	26.3	14.6
Firms		52,139			5,397			46,742	
Firm-years		554,571			75,821			478,750	

Firms with access to foreign credit are those that borrow in foreign currency at some point in the sample. Levels are in Rs. million and ratios in percent. For firms with access to foreign credit, the bank-debt and FI/NBFC shares of borrowings are computed over the firm-years that carry foreign-currency debt; other statistics use every firm-year. Ratios are winsorized at the 1st and 99th percentiles for means and standard deviations; medians are raw. Differences between the two groups are significant at the 1% level.

$$y_{i,t+h} = \alpha_i + \delta_{jt} + \beta\big(\text{MaPP}_t \times D_i\big) + \gamma\big(\text{MaPP}_t \times X_i\big) + \Lambda' Z_{i,t-1} + \varepsilon_{i,t+h}, \qquad (1)$$

Where y_{it+h} is the borrowing outcome for firm i in year $t+h$, MaPP_t is the net change in the macroprudential index over the firm's fiscal year t which is April to March. D_i is the firm's initial bank-debt-to-assets ratio, standardized. The term X_i contains the firm's initial size, leverage and exporter status, each interacted with the shock. This ensures that β is isolated from the differences in firm size, leverage and exporting. Z_{it-1} is a set of firm level controls namely size, leverage, profitability, tangible capital and lagged dependent variable. α_i captures all fixed differences across firms and δ_{jt} absorbs conditions common to all firms in the same industry j in a year therefore capturing the main effect of $MaPP_t$ as well as any shock specific to an industry in that year, such as a sector specific change in bank lending. The coefficient of interest is β. It measures how much more borrowing of a bank-dependent firm adjusts after tightening, in

Table 2: Foreign-currency debt intensity among firms with access to foreign credit

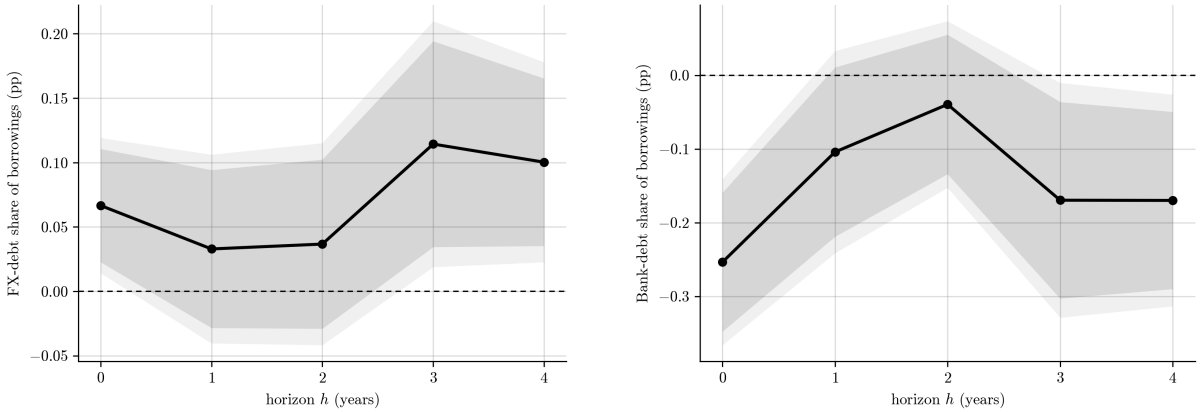
Statistic	Mean	Median	p25	p75
Share of firm-years with foreign-currency debt > 0 (%)	33.9			
Foreign-currency debt share of total borrowings (%)	38.8	29.0	11.3	62.2
Foreign-currency debt / assets (%)	15.1	9.1	3.5	19.5
Export earnings / sales (%)	19.3	2.0		
Firms with zero export revenue (%)	43.0			
Net foreign-currency short: FX debt exceeds annual exports (%)	64.6			
Firms holding foreign-currency debt		5,397 firms		

Rows 2 to 6 are conditional on holding foreign-currency debt. "Net foreign-currency short" means foreign-currency debt exceeds the firm's annual export earnings. Shares are winsorized at the 1st and 99th percentiles for means; medians and percentiles are raw.

percentage points of the outcome for a one standard deviation increase in initial bank dependence and a one unit tightening of the index relative to a firm with zero initial bank debt. Standard errors are clustered over 3-digit industries (about 120 clusters) in the baseline specification. Since δ_{jt} removes everything common to an industry in a year, the dependence left in the errors runs within industries over time, which clustering over industries allows for. I further do robustness tests using clustering over years (at most 19 clusters), and over industries and years.

3.2 Firm's Debt Composition

Figure 2 plots how the composition of firm debt responds to a macroprudential tightening. The left panel shows the share of a firm's total borrowing that is in foreign currency and the right panel shows the share from banks. After a tightening, more bank dependent firms raise the foreign currency share of their borrowing and reduce the bank share. The foreign currency share rises by around 0.08 percentage points following the tightening and by about 0.11 percentage points three years out. Bank debt moves the other way and it moves first, falling by about 0.24 percentage points following a macroprudential tightening.



(a) Foreign Currency Debt Share

(b) Bank Debt Share

Figure 2: Response of Debt Composition to Macroprudential Tightening

Notes: The figures plot impulse responses of the foreign-currency debt share (a) and bank debt share (b) following a macroprudential tightening shock. All regressions include firm and 3-digit industry by year fixed effects, and standard errors are clustered by industry. Darker shaded regions denote 90% and lighter shaded regions 95% confidence intervals.

Total borrowing however, barely moves as shown in Figure 3. Across all horizons, the response of total borrowing, is close to zero and never statistically different from zero. This is consistent with firms having their financing needs tied to investment plans set well in advance, carried out over a long horizon. Consequently, it is costly for a firm in the middle of a planned project to contract borrowing when bank lending tightens. So it seeks to replace debt rather than forego profitable investments.

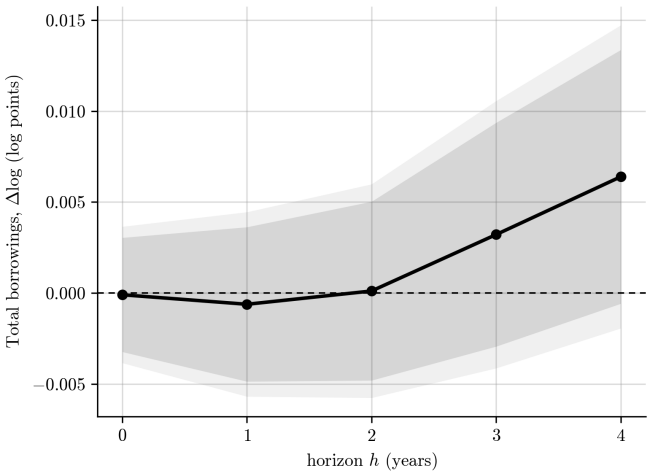


Figure 3: Response of Total Borrowings to a Macroprudential Tightening

Notes: The figure plots the impulse response of total borrowings, using the cumulative log change in total borrowings from t to $t + h$. All regressions include firm and 3-digit industry by year fixed effects, and standard errors are clustered by industry. Darker shaded regions denote 90% and lighter shaded regions 95% confidence intervals.

Substitution Away from Banks: A natural alternative source of financing for firms are non-bank financial corporations (NBFC) which lent heavily to firms in years when banks were weighed down by non-performing assets. I estimate the same specification with the NBFC share of borrowing as the outcome variable in figure 4. The NBFC share of borrowing increases by about 0.25 percentage points over the first three years before fading. Together with the foreign currency debt share, this roughly offsets the drop in the bank share following a macroprudential tightening. Non bank credit, however, is a small part of these firms' borrowing. It accounts for the smallest share of debt at about 8 percent compared to almost 30 percent for foreign debt and nearly half for bank debt. More importantly, non-bank credit stays in rupees. The shift into foreign currency is the part of the substitution that changes the risk on firm balance sheets.

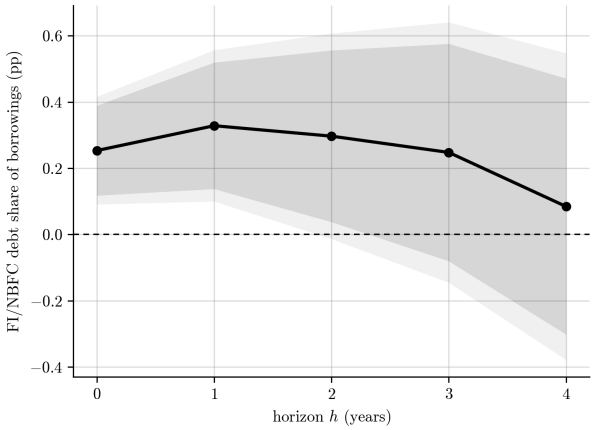
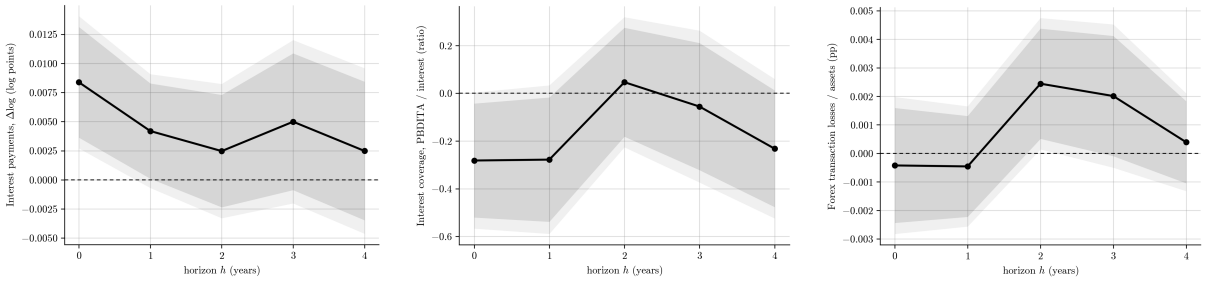


Figure 4: Debt from Non-Bank Financial Corporations (NBFC)

Notes: This figure plots the impulse responses of firms' NBFC debt share following a macroprudential tightening shock. All regressions include firm and industry by year fixed effects, and standard errors are clustered by industry. Darker shaded regions denote 90% and lighter shaded regions 95% confidence intervals.

Costs of Substitution: The cost of substitution shows up on the income statement of firms. Figure 5 plots the response of interest payments and interest coverage of firms, measured by the operating profit over interest expense. This represents how many times over a firm's earnings could pay its interest bill. As the ratio falls, a larger share of earnings goes into servicing its debt. After a macroprudential tightening, interest coverage falls while the interest payments rise concurrently. Put together, these firms are left with a heavier interest burden relative to what they earn. Foreign exchange transaction losses relative to assets also rise two years after the tightening. This is consistent with the early sign that the foreign currency debt the firms have taken on exposes them to exchange rate swings.



(a) Interest Payments (b) Interest Coverage (c) Forex Losses / Assets

Figure 5: Costs of Substitution

Notes: The figures plot the impulse responses of the log changes in interest payments (a), interest coverage ratio measured by PBDITA over interest (b) and forex transaction losses as a ratio of firm assets (c). All regressions include firm and industry by year fixed effects, and standard errors are clustered by industry. Darker shaded regions denote 90% and lighter shaded regions 95% confidence intervals.

3.3 Exposure to Exchange Rate Fluctuations

The substitution towards foreign currency debt leaves firms more exposed to exchange rate fluctuations. From 2003 to 2022, the rupee has depreciated against the dollar with sharp episodes during the Global Financial Crisis, Taper Tantrum, and the recent COVID-19 Pandemic and following US rate hikes. Close to half the firms carrying foreign-currency debt report no export revenue and about two thirds owe more in foreign currency than they earn from exports in a year. Since majority of the depreciation episodes are concentrated in these few years it is diluted when outcomes are averaged over all the years after a macroprudential tightening. A single tightening also moves any one firm's exposure only a little. A test that asks directly whether the exposure created by policy hurts firms during a depreciation therefore has little power. [10] I therefore, split the argument into two steps. First from macroprudential policy to greater exposure which is the substitution result in the preceeding section. Second, from exposure to harm where I ask whether firms that carry more foreign-currency debt do worse when the rupee depreciates, using all the variation in exposure across firms, not only the part created by the policy. Neither link alone shows that a tightening leaves firms more fragile, but together they do, as long as the foreign-currency debt taken on after a tightening is exposed to the rupee in the same way as any other. I estimate:

[10] I also estimate this direct version, adding to the local projection a triple interaction of the shock, bank dependence, and the depreciation in the outcome year. The estimates are imprecise and change sign across horizons, as expected when a small policy-induced change in exposure is multiplied by the change in exchange rate.

$$y_{i,t} = \gamma\left(\text{FX}_{i,t-1} \times \text{Dep}_t\right) + \theta \cdot \text{FX}_{i,t-1} + \phi \log A_{i,t} + \alpha_i + \delta_{jt} + \varepsilon_{i,t} \tag{2}$$

Where $\text{FX}_{i,t-1}$ is the firms foreign-currency borrowing relative to assets at the end of the previous fiscal year, measured in percent. Dep_t is the depreciation of the rupee over the firms fiscal year. A_{it} is total assets and I also have firm and industry by year fixed effects represented by α_i and δ_{jt} respectively. The interaction terms are divided by 10 so that γ is the effect on a firm with 10 percentage points more foreign currency debt relative to assets in a year when the rupee falls by 10 percent and is interpreted as the cost of carrying foreign currency exposure. 10 percentage points of assets is close to the median and the rupee fell by 10 percent or more during major episodes of downturns such as 2008 financial crisis, 2011-12 during the Eurozone crisis, 2013 during the Taper Tantrum, 2020 during COVID outbreak and 2022 when the US raised interest rates aggressively.

Figure 6 shows that the exposure is indeed costly for firms. Forex losses rise by about 0.11 percentage points of assets, a little over half of their sample mean. Operating profits and return on assets fall as well. In the sample, operating earnings fall short of interest payments in about 16.1 percent of firm-years. For such a firm, a depreciation of 10 percent raises that share by 1.2 percentage points to 17.3 percent. The share of firm-years with a loss rises from 25.6 percent to 27.4 percent. In both cases, the distress becomes roughly 7 percent more likely than it would otherwise be.

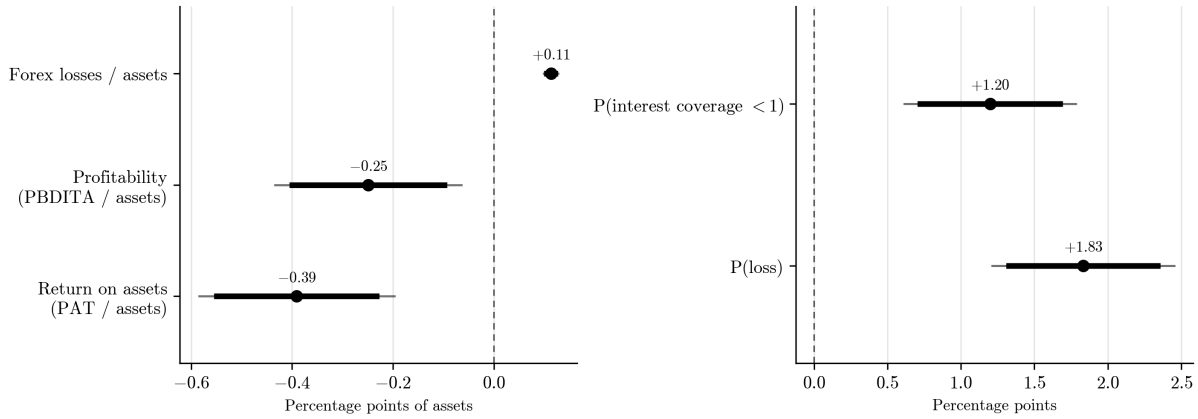


Figure 6: Firm losses when Rupee depreciates

Notes: Each point is the coefficient γ for one outcome, per 10 percentage points of foreign-currency debt to assets times 10 percent fiscal-year depreciation of the rupee. The left panel shows forex losses, operating profitability (PBDITA over assets), and return on assets (profit after tax over assets), in percentage points of assets. The right panel shows the probability that operating earnings fall short of interest expense and the probability of a loss, in percentage points. All regressions include firm and industry by year fixed effects, and standard errors are clustered by industry. Thick bars denote 90% and thin bars 95% confidence intervals.

Taken together with the previous results, the evidence links macroprudential policy to firm fragility in two steps. A tightening pushes bank-dependent firms toward foreign-currency debt, and foreign-currency debt raises losses and the risk of distress when the rupee depreciates.

3.4 Robustness

Standard errors: For the baseline results, I cluster standard errors by industry. The main threat to inference is correlation within a firm over time since firm balance sheets are persistent. Firms in the same industry also face the same demand, costs and lending conditions, so their errors may move together. Since each firm belongs to a single industry, clustering by industry adjusts for arbitrary correlation of both kinds, within a year and across years. Additionally, industry by year fixed effects absorb any part of the error that is common to all firms in an industry in a year and when a common shock affects firms in a sector alike, these dummies account for it. Some correlation across firms within a year may still remain if firms in different industries but with similar exposure to the policy shock respond similarly for reasons not captured by the fixed effects and industry clustering. This is a narrower concern. The most natural reason

for similarly bank dependent firms to respond alike to a macroprudential tightening is that they are in the same sector and face the same conditions, which the baseline already allows for. I nonetheless report standard errors clustered by year and two-way clustered by industry and year in Figure A.1. Both rely on the time dimension, and our sample spans at most twenty fiscal years. Cluster-robust inference with so few clusters is unreliable (Bertrand et al. 2004), so I treat these as demanding robustness checks. They change only the precision of the estimates, not the estimates themselves. The two sources firms turn to after a tightening both hold up. The rise in borrowing from non-bank financial institutions remains significant one and two years out under both alternatives. The rise in the foreign currency share also remains significant three and four years out under both. The impact responses and the fall in the bank share keep their sign and size but lose precision.

Borrower-based instruments: The results do not depend on whethe the index includes borrower-based instruments. The baseline index drops borrower-based tools and captures only the lender-side tightening. It may be of concern that certain borrower-based instruments such as loan-to-value limits, debt to income caps and debt service to income limits directly constrain how much credit domestic firms can obtain from local financial institutions. When domestic regulations cap a firm's local borrowing capacity, the firm faces a similar incentive to bypass these domestic limits by tapping into foreign credit markets. Thus, in certain cases, borrower based constraints can trigger the same leakage as lender based instruments. Rebuilding the index with borrower based instrument: DTI, LTV and DSTI included leaves the estimates, plotted in the appendix figure A.2 very close to the baseline, if slightly smaller on impact.

Monetary Policy: Macroprudential policy may coincide with monetary tightening which could itself push firms abroad. I add the monetary policy rate, interacted with bank dependence alongside the macroprudential term and plot the responses in figure A.3 in the appendix. The macroprudential coefficient is essentially unchanged. The foreign currency share still rises and the bank share falls by nearly the same amounts, so the substitution is a response to the macroprudential policy and not due to any correlated monetary stance.

Intensity-based Measure: The macroprudential index used in the analysis counts the net tightening actions thereby giving us an aggregate stance. In doing so, every macro-

prudential action is treated the same. So all tightenings are treated equally irrespective of the intensity of tightening. To capture the intensive margin, I build a second measure of shock from the actual levels of India's two main quantity instruments, the reserve requirements and the capital requirements. The shock is the change in these levels over the firm's fiscal year, standardized. The baseline results hold and is in fact, stronger as shown in figure A.4 in the appendix. The foreign currency share increases across every horizon and the bank share falls at every horizon while the total borrowing stays flat and insignificant.

4 Model

The empirical results in the previous section show that a macroprudential tightening moves bank-dependent firms out of bank credit and into foreign-currency debt, and that debt leads to distress when the currency depreciates. This is one side of the policy trade-off. The other side is the reason that the policy exists. A better capitalized banking system is less likely to be insolvent during a downturn and can keep lending when firms need credit the most. This benefit works through the banking system as a whole and is shared by all firms, so it calls for a framework that looks at the economy as a whole. The cost is also larger than the firm-level estimates suggest, because firms that default pass on a part of the loss back to banks which the policy was meant to protect. Weighing the two requires careful consideration of both channels. To this extent, in this section I set up a two-period small open economy model in which firms borrow from domestic banks and from foreign lenders and a macroprudential policy tightening in the form of capital requirement on banks raises the cost of domestic debt. I use the model to show how the regulator trades off the credit which a tightening preserves in a crisis against the credit it forgoes and the fragility it creates during normal times, and how this trade-off shapes the optimal level of the requirement.

4.1 Environment

I consider a two-period economy, $t = 1, 2$, with a firm or entrepreneurial sector and a bank or financial sector. A continuum of firms borrow from domestic banks and from foreign lenders to buy capital and produce a final good. A continuum of banks raise deposits from households who supply them elastically at a gross real rate R and lend to firms. Foreign lenders supply funds denominated in foreign currency elastically at

a gross real rate R^* to firms.

4.1.1 Aggregate shocks

At the end of each period an aggregate state $\omega = \{e', A\}$ realizes where e' is the end of period exchange rate (domestic units per \$) [11] and A is an aggregate productivity shock. For simplicity, I assume both are two-point and have mean = 1:

- $e' \in \{e_L, e_H\}$ built from log-shocks $+/-\sigma_e$ rescaled so $E[e'] = 1$ exactly.
- $A \in \{1-\sigma_A, 1+\sigma_A\}$, so $E[A] = 1$ exactly.

I assume that the correlation between the two shocks is 0. Shocks are i.i.d across periods so the realized ω enters the future only through the net-worth stocks it leaves behind; it is not itself a state variable.

4.1.2 Idiosyncratic Shock

Each firm draws an idiosyncratic productivity shock $z \sim \text{LogNormal}(\mu_z, \sigma_z^2)$, with $\mathbb{E}[z] = 1$, which implies $\ln z \sim \mathcal{N}(\mu_z, \sigma_z^2)$ and $\mu_z = -\frac{\sigma_z^2}{2}$.

$\Phi_N(\cdot)$ is the cumulative distribution function (CDF) of the standard normal distribution. I define $\bar{z}$ as the threshold value of z below which the firm falls short of its debt obligations and defaults for a given state ω. Then the probability of firm's default can be defined as:

$$F(\bar{z}) = \Phi_N\left(\frac{\ln \bar{z} - \mu_z}{\sigma_z}\right) \tag{3}$$

And the expected productivity over the default region is given by:

$$G(\bar{z}) = \int_0^{\bar{z}} z\, dF(z) = \Phi_N\left(\frac{\ln \bar{z} - \mu_z - \sigma_z^2}{\sigma_z}\right) \tag{4}$$

Following Bernanke et al. (1999), I define the fraction of the project value accruing to lenders before default costs as

$$\Gamma(\bar{z}) = G(\bar{z}) + \bar{z}\big(1 - F(\bar{z})\big) \tag{5}$$

And the fraction retained by firms is therefore:

$$1 - \Gamma(\bar{z}) = \int_{\bar{z}}^{\infty} (z - \bar{z})\, dF(z) \geq 0 \tag{6}$$

[11] so a rise corresponds to depreciation

4.2 The Firm

4.2.1 Technology and balance sheet

Each firm enters the period with net worth n^F. It uses the net worth, together with debt, to acquire capital k. The firm can borrow in two ways: from domestic banks in domestic currency, b_d, and from foreign lenders in foreign currency, b_f. It makes these choices at the start of the period, before the aggregate state ω and its own productivity z are realized.

At the end of the period, a firm with capital k and productivity z earns $zAR^K k$ in state ω, where R^K is the gross return per unit of capital and depends on the total stock of capital at the aggregate level. The payoffs are proportional to the firms' individual capital and so each firm operates with constant returns to scale. Further, since each firm is small, they take R^K as given.

The firm finances its capital with its net worth and debt. Foreign debt is denominated in foreign currency, so its value in domestic units depends on the exchange rate e at which it is taken on. The balance sheet of the firm is

$$k = n^F + b_d + e\, b_f, \tag{7}$$

so total debt, measured in domestic units, is $b_d + e\, b_f = k - n^F$. I normalize the exchange rate at the time of borrowing to $e = 1$.

The share of domestic currency debt is given by,

$$s = \frac{b_d}{b_d + e\, b_f}, \tag{8}$$

so that $1 - s$ is the share of foreign currency debt. The firm's financing choice can then be summarized by two numbers: how much it borrows in total, and how it splits that borrowing between the two currencies.

The split matters because the two debts are repaid at different exchange rates. Domestic debt is repaid in domestic currency, so its burden does not depend on the exchange rate. Foreign debt is repaid at the end-of-period rate e'. If the currency depreciates, so that $e' > e$, the firm owes more in domestic units on its foreign debt. This is the source of the firm's currency risk.

4.2.2 Entry and exit

At the end of period 1, a fraction τ_x^F of firms exit. An exiting firm pays out its wealth and consumes it. New firms enter with total net worth equal to a fraction ξ^F of aggregate firm net worth N^F. Since firms' choices do not depend on their size (as shown below), only aggregate firm net worth matters, not the number of firms.

This assumption plays two roles. First, it keeps firm net worth stationary. Period 1 represents the steady state, and without exit, firms would carry all of their earnings into period 2, and firm net worth would grow even if no shock occurred. With exit and entry, any difference between period 2 and the steady state comes from the aggregate shock alone. Banks face the same assumption. Second, it determines how much the firm cares about period 2. With probability τ_x^F the firm exits and consumes its wealth at the end of period 1, and only with probability $1 - \tau_x^F$ does that wealth stay in the firm and earn a return in period 2. The exit rate therefore sets the weight the firm places on period 2 when it makes its choices in period 1. Entry does not affect this weight, since the net worth of entrants goes to new firms.

4.2.3 Default threshold

The contractual interest rates that the firm faces on both kind of debts are fixed when it borrows. Let R_d and R_f denote the gross rates on domestic and foreign debt. At the end of the period, after the shocks are realized, the firm owes $R_d b_d$ on its domestic debt and $e' R_f b_f$ on its foreign debt. Its total debt service in domestic units is therefore,

$$R_d \, b_d + e' R_f \, b_f.$$

The firm repays if its project value $zAR^K k$ covers this amount and defaults otherwise. It therefore defaults when its productivity falls below the threshold

$$\bar{z}(\omega) = \frac{R_d \, b_d + e' R_f \, b_f}{AR^K k}. \tag{9}$$

The threshold depends on the aggregate state through e' and A. A depreciation raises e' and with it the domestic value of the foreign debt, while low productivity lowers the project value. Both raise $\bar{z}$, so more firms default. Since the cutoff is homogenous of degree 0, the threshold depends on the proportions of the balance sheet

and not the size.[12]

4.2.4 Lenders

Lenders are risk neutral and competitive. Domestic banks fund their loans at the cost $R(1+\tau)$, where τ is the macroprudential policy wedge created by the capital requirement. I take τ as given here and derive it from the banks' problem in the subsequent section. Foreign lenders fund their loans at the world rate R^*.

If a firm repays, each lender receives what they are owed. If a firm defaults, its lenders recover a fraction κ of the project value, and this amount is shared between them in proportion to their claims. Foreign lenders, however, keep only a fraction $\varepsilon(s)$ of their share which falls as the firm borrows more abroad. So they recover less of each additional unit they lend. Foreign creditors are likely to face higher costs when recovering their claims from distressed firms in India. Domestic creditors tend to know the local legal system better and have greater access to information about the borrower and its assets. This gives them an advantage in monitoring borrowers, negotiating restructuring and enforcing their claims, which can lead to higher recovery. In contrast, foreign creditors may find it harder to navigate India's slow-moving and complex judicial and insolvency resolution system.[13] This friction acts only after a default, so it is an ex-post recovery friction, not an ex-ante monitoring cost.

$$\varepsilon(s) = 1 - (1-s)^{\psi}, \qquad \psi > 1.$$

The more exposed a foreign lender is to a firm, the higher contractual rate they charge because they are likely to recover less. Since $\psi > 1$, the rising marginal cost of foreign debt keeps the firms from moving entirely to foreign debt.

Per unit of debt owed, the expected repayment to each type of lender is

$$\varphi^d(\bar{z}) = \big(1 - F(\bar{z})\big) + \kappa \frac{G(\bar{z})}{\bar{z}}, \qquad \varphi^f(\bar{z}, s) = \big(1 - F(\bar{z})\big) + \varepsilon(s)\, \kappa \frac{G(\bar{z})}{\bar{z}}.$$

The first term is full repayment, received when the firm survives. The second term is

[12] Proof in Appendix B.1

[13] More broadly, $\varepsilon(s)$ is a reduced-form way of capturing anything that makes foreign credit costlier or harder to acquire than domestic credit as a firm borrows more abroad, such as hedging costs, rollover risk, information frictions or regulatory limits on external commercial borrowing. Firms with access to foreign credit hold both domestic and foreign debt rather than borrowing only in the cheaper market. This requires the cost of foreign credit to rise with firm's foreign borrowing, which $\varepsilon(s)$ delivers.

the recovery when the firm defaults.[14]

The interest rates are set before the shock, so lenders are compensated to the point where they break even in expectation across the aggregate states.

$$R_d \, \mathbb{E}_\omega \left[\varphi^d(\bar{z}(\omega)) \right] = R(1+\tau), \qquad R_f \, \mathbb{E}_\omega \left[\varphi^f(\bar{z}(\omega), s) \right] = R^*. \tag{10}$$

The exchange rate does not appear in the foreign lenders' condition. They lend and are repaid in foreign currency, so a depreciation does not change the value of what a surviving firm pays them. It matters to them only because it makes default more likely, through $\bar{z}(\omega)$. Given the firm's leverage ($\lambda = k/n^F$) and debt mix (s), these two conditions determine the interest rates R_d and R_f.

4.2.5 The firm's problem

The firm makes its choices at the start of the period, before the shocks are known. Given its net worth n^F, it chooses its domestic and foreign debt, or equivalently, the leverage $\lambda \equiv k/n^F$ and domestic debt share s. These two numbers pin down its capital and its two debts:

$$k = \lambda \, n^F, \qquad b_d = s(\lambda - 1) \, n^F, \qquad b_f = (1-s)(\lambda - 1) \, n^F.$$

The interest rates R_d and R_f are not fixed and are set by the lenders' conditions as outlined in Section 4.2.4, so they change with the firm's choices: a firm that borrows more is riskier and pays more. The firm takes this into account when it chooses.

After the shocks, a firm with productivity z keeps what is left after paying its debt, or nothing if it defaults.

$$\max\{zAR^K k - (R_d b_d + e' R_f b_f), 0\}$$

Averaging over the firm's own idiosyncratic draws, its wealth at the end of the period in state ω is

$$W^F(\omega) = A(\omega) \, R^K k \left[1 - \Gamma(\bar{z}(\omega)) \right], \tag{11}$$

the share of the project value that remains with the firm. Limited liability is built into Γ: a firm that defaults keeps nothing, but never less than nothing. The firm then

[14] These are the BGG expressions written directly. In the notation of Bernanke et al. (1999), $\varphi^d = \left[\Gamma(\bar{z}) - (1-\kappa) G(\bar{z}) \right] / \bar{z}$, and similarly for φ^f with $\varepsilon(s)\kappa$ in place of κ.

either exits with probability τ_x^F and consumes its wealth or it keeps its wealth as net worth for the next period. The firm's problem in period 1 is

$$V_1^F(n^F, S) = \max_{\lambda, s} \frac{1}{R} \mathbb{E}_\omega \left[\tau_x^F W^F(\omega) + (1 - \tau_x^F) V_2^F \left(W^F(\omega), S'(\omega) \right) \right] - n^F \Phi(\lambda), \quad (12)$$

subject to the two lenders' conditions in equation 10 and the balance sheet identity in equation 7.[15] Here V_2^F is the value of a firm that enters period 2 with net worth n^F. Period 2 is the last period, so the firm consumes all of its wealth at the end of it:

$$V_2^F(n^F, S') = \max_{\lambda', s'} \frac{1}{R} \mathbb{E}_{\omega'} \left[W^{F\prime}(\omega') \right] - n^F \Phi(\lambda'), \quad (13)$$

where ω' is the aggregate state realized at the end of period 2, and $W^{F\prime}(\omega')$ is end-of-period wealth, defined as in period 1 but with the period-2 choices, the wedge τ' and the return on capital $R^{K\prime}$. The aggregate state is $S \equiv (N^F, N^B)$, aggregate firm and bank net worth. This matters to the firm because it determines the credit wedge and the return on capital. The shock ω is not part of the state. Shocks are independent over time, so once ω has hit balance sheets, it affects the future only through the net worth it leaves behind, $S'(\omega)$.

In the period-1 problem, the first term is the wealth consumed if the firm exits. The second is the value of continuing with that wealth as net worth. The last term is a leverage adjustment cost,

$$\Phi(\lambda) = \frac{\chi}{2} \frac{(\lambda - \bar{\lambda})^2}{\bar{\lambda}},$$

which grows as leverage moves away from a reference level $\bar{\lambda}$. The cost captures the idea that firms have a target capital structure and adjust toward it slowly. It limits how much total borrowing responds to the wedge τ, in line with the evidence that firms adjust the mix of their debt more than its size. I set $\bar{\lambda}$ equal to the leverage that firms choose under the current existing capital requirement. At $\lambda = \bar{\lambda}$ the cost and slope is zero and has no effect on firms' choices under the existing policy.

The firm's size does not affect its choices. Technology has constant returns, the balance sheet scales with the firm, the lenders' conditions is homogeneous of degree 0 and depend only on the ratios λ and s, and the adjustment cost is proportional to net worth. A firm with twice the net worth can therefore double every quantity, pay the

[15] Formally, the expectation runs over both the aggregate state ω and the firm's own draw z. Because the value functions are linear in net worth (shown below), only the firm's average wealth over z matters, which is $W^F(\omega)$.

same interest rates, and end each period with twice the wealth. In period 2, which has no continuation, its value is then proportional to its net worth, $V_2^F(n^F, S') = \nu_2^F(S')\, n^F$, where $\nu_2^F(S')$ is the value of one unit of net worth in period 2:

$$\nu_2^F(S') = \max_{\lambda', s'} \frac{1}{R}\, \mathbb{E}_{\omega'} \Big[A(\omega') R^{K\prime}\, \lambda' \left(1 - \Gamma(\bar{z}'(\omega'))\right) \Big] - \Phi(\lambda'). \tag{14}$$

Substituting this into the period-1 problem, the same argument gives $V_1^F(n^F, S) = \nu_1^F(S)\, n^F$, with

$$\nu_1^F(S) = \max_{\lambda, s} \frac{1}{R}\, \mathbb{E}_{\omega} \Big[m(\omega)\, A(\omega) R^K(S)\, \lambda \left(1 - \Gamma(\bar{z}(\omega))\right) \Big] - \Phi(\lambda), \tag{15}$$

$$\text{where, } m(\omega) = \tau_x^F + (1 - \tau_x^F)\, \nu_2^F\big(S'(\omega)\big). \tag{16}$$

The weight $m(\omega)$ is the value of one unit of wealth at the end of the period in state ω. With probability τ_x^F it is consumed and worth one. With probability $1 - \tau_x^F$ it stays in the firm, where it is worth $\nu_2^F(S'(\omega))$. A unit of net worth is worth more than one inside the firm, because it supports more than one unit of capital. How much more depends on credit conditions in period 2, which differ across states. This is what makes the firm forward-looking: it cares not only about how much wealth it ends with, but about which state that wealth arrives in. The period-2 problem is the same problem with $m = 1$ in every state, since all wealth is then consumed. Setting $m = 1$ in period 1 as well gives the static problem of a myopic firm.

Because all firms choose the same leverage and debt mix, aggregate firm net worth at the start of period 2 is

$$N^{F\prime}(\omega) = \Big[(1 - \tau_x^F)\, A(\omega) R^K \lambda \left(1 - \Gamma(\bar{z}(\omega))\right) + \xi^F \Big] N^F. \tag{17}$$

The first term is the wealth of surviving firms and the second is the net worth of entrants. This is the firm part of next period's state $S'(\omega)$.

4.2.6 Optimality conditions

The firm has two choices, the composition of debt s and leverage λ, so there are two optimality conditions. The cost Φ depends only on leverage, so it appears in the leverage condition and not in the composition condition. The results on currency substitution, which is the main focus of the paper, therefore do not depend on the form of Φ.

Debt composition. Raising s while holding λ fixed replaces a unit of foreign debt with a unit of domestic debt, leaving total debt and capital unchanged. [16] The default threshold can be written as $\bar{z}(\omega) = \frac{\lambda-1}{\lambda} \cdot \frac{D(\omega)}{AR^K}$, where $D(\omega) \equiv R_d\, s + R_f\, e'(1-s)$ is debt service per unit of debt. The composition condition is

$$\mathbb{E}_\omega \Big[m(\omega) \left(1 - F(\bar{z}(\omega))\right) \frac{\partial D(\omega)}{\partial s} \Big] = 0. \tag{18}$$

The firm shifts its debt mix until a weighted average of the change in debt service is zero. Each state is weighted by the value of wealth there, $m(\omega)$, and by the probability that the firm survives, $1 - F(\bar{z}(\omega))$.

This condition has a simple reading. Let $\mathcal{D} \equiv R_d + s\, \partial R_d / \partial s$ be the marginal cost of domestic borrowing and $\mathcal{F} \equiv R_f - (1-s)\, \partial R_f / \partial s$ the marginal cost of foreign borrowing. Each of these marginal costs is the rate on the extra unit plus the change in the rate faced by the debt the firm already has. The condition in equation 18 becomes

$$\mathcal{D} = \hat{e}\, \mathcal{F}, \qquad \hat{e} \equiv \frac{\mathbb{E}_\omega \big[m(\omega) \left(1 - F(\bar{z}(\omega))\right) e' \big]}{\mathbb{E}_\omega \big[m(\omega) \left(1 - F(\bar{z}(\omega))\right) \big]}. \tag{19}$$

The firm equates the marginal costs of the two debts, converting the foreign cost at $\hat{e}$, a weighted average of the end-of-period exchange rate. Because default is most likely after a depreciation, the survival weights are lowest in exactly those states. The firm therefore puts too little weight on the states where foreign debt is most expensive, and $\hat{e}$ lies below the true expectation $\mathbb{E}[e'] = 1$. Limited liability makes foreign debt look cheaper to the firm than it is.

Using the lenders' conditions to replace $\partial R_d / \partial s$ and $\partial R_f / \partial s$ gives the condition in terms of the price gap:

$$\underbrace{R_d - \hat{e}\, R_f}_{\text{price gap}} = \underbrace{(1-s)\, \hat{e}\, \frac{R_f}{\mathbb{E}[\varphi^f]}\, \varepsilon'(s)\, \mathbb{E}\left[\varphi^f_\varepsilon\right]}_{\text{marginal cost of foreign debt}} + \underbrace{s\, \frac{R_d}{\mathbb{E}[\varphi^d]}\, \mathbb{E}\left[\varphi^d_{\bar{z}}\, \frac{\partial \bar{z}}{\partial s}\right] + (1-s)\, \hat{e}\, \frac{R_f}{\mathbb{E}[\varphi^f]}\, \mathbb{E}\left[\varphi^f_{\bar{z}}\, \frac{\partial \bar{z}}{\partial s}\right]}_{\text{repricing of default risk}} \tag{20}$$

where $\varphi^f_\varepsilon = \kappa G(\bar{z}) / \bar{z}$ is the effect of ε on foreign recovery and $\varphi_{\bar{z}}$ denotes the derivative of φ with respect to the threshold. The left side is the interest saving from borrowing one more unit in foreign currency. The right side is what that unit costs: there is a worsening in the terms of the loan when a firm becomes more dollarized and face a

[16] In a formulation with b_d and b_f as choice variables, this condition is the difference between the two first-order conditions.

steeper price, and the change in default risk reprices both debts.[17]

Leverage. The leverage condition is

$$\frac{1}{R}\mathbb{E}_\omega\left[m(\omega)\left\{\underbrace{A(\omega)R^K(1-\Gamma(\bar{z}))}_{\text{return on one more unit of capital}} - \underbrace{(1-F(\bar{z}))\left[\frac{D(\omega)}{\lambda}+(\lambda-1)\frac{\partial D(\omega)}{\partial\lambda}\right]}_{\text{extra debt service, incl. repricing}}\right\}\right] = \Phi'(\lambda) \tag{21}$$

The left side is the value of raising leverage by one unit: the extra capital earns its return, and the extra borrowing raises debt service, including the higher rate on existing debt, in the states where the firm survives. The right side is the marginal cost of adjusting leverage.

How policy enters. The composition condition shows how a tighter capital requirement changes the debt mix. A higher requirement raises the wedge τ, which raises the domestic rate R_d through the domestic lenders' condition and widens the price gap. The firm shifts toward foreign debt until the marginal costs are equal again. This raises its exposure to the exchange rate, so default becomes more likely after a depreciation. The firms take the wedge τ as given. The next subsection turns to the banks, where I show where it comes from.

4.3 The Bank

In the lenders' subsection, domestic banks require an expected return of $R(1+\tau)$ on their loans. This section describes the banks and shows where the wedge τ comes from.

4.3.1 Balance sheet and returns

A bank enters the period with net worth n^B, lends b_d to firms, and funds the rest with deposits $b_d - n^B$ at the gross rate R. Banks lend only in domestic currency, so they carry no exchange rate exposure of their own. Each bank is subject to a capital requirement,

$$n^B \geq \theta\, b_d, \tag{22}$$

[17] With a single state and $m = 1$, the firm minimizes its default threshold, $\partial\bar{z}/\partial s = 0$, the repricing terms vanish, and the condition reduces to the myopic firm's optimality condition.

which limits how much it can lend per unit of its own net worth. The requirement θ is the macroprudential policy instrument: a tighter policy means a higher θ.

Banks are competitive and take the return on loans as given. Per unit lent, a bank receives R_d from firms that repay and a recovery from firms that default:

$$\rho^B(\omega) = \underbrace{R_d\left(1 - F(\bar{z}(\omega))\right)}_{\text{repayment by surviving firms}} + \underbrace{R_d \cdot \kappa \frac{G(\bar{z}(\omega))}{\bar{z}(\omega)}}_{\text{recovery from defaulting firms}} \tag{23}$$

By the domestic lenders' condition in equation 10, the expected return on a loan is $\mathbb{E}_\omega[\rho^B(\omega)] = R_d\,\mathbb{E}_\omega[\varphi^d(\bar{z}(\omega))] = R(1+\tau)$.

The bank's assets earn $\rho^B(\omega)$ and its deposits cost R, so its wealth at the end of the period is

$$W^B(\omega) = \rho^B(\omega)\, b_d - R\,(b_d - n^B) = b_d\big(\rho^B(\omega) - R\big) + R\, n^B. \tag{24}$$

Deposits are insured, so depositors are repaid in full even if the bank fails.

4.3.2 Exit and entry

Banks face the same structure of exit and entry as firms. At the end of period 1, a fraction τ_x^B of bankers exit and consume their wealth. New bankers enter with total net worth equal to a fraction ξ^B of aggregate bank net worth in period 1, N^B. In the aggregate, this inflow is equivalent to an outside recapitalization of the banking system, and I calibrate it to the equity injections into Indian banks.

Without exit, bankers would retain all of their earnings, and bank net worth would grow even if no shock occurred. The entry and exit of banks also helps pin down the return on bank equity and the wedge τ as I show below. Like in firms, it sets the weight the banker places on period 2. Entry does not affect this weight, since the inflow goes to new bankers, not to existing ones.

4.3.3 The bank's problem

Let $\lambda^B \equiv b_d/n^B$ denote bank leverage, so the capital requirement becomes $\lambda^B \leq 1/\theta$. A bank chooses its leverage in period 1 to solve

$$V_1^B(n^B, S) = \max_{\lambda^B \leq 1/\theta} \frac{1}{R}\,\mathbb{E}_\omega\Big[\tau_x^B W^B(\omega) + (1-\tau_x^B)\, V_2^B\big(W^B(\omega),\, S'(\omega)\big)\Big] \tag{25}$$

Here V_2^B is the value of a bank that enters period 2 with net worth n^B. Period 2 is the last period, so the bank consumes all of its wealth at the end of it:

$$V_2^B(n^B, S') = \max_{\lambda^{B\prime} \leq 1/\theta} \frac{1}{R} \mathbb{E}_{\omega'} \left[W^{B\prime}(\omega') \right], \tag{26}$$

where $W^{B\prime}(\omega')$ is end-of-period wealth, defined as in period 1 but with period-2 leverage, and the period-2 return on loans.

The bank's size does not affect its choices. A bank lends to many firms, so its return per unit lent, $\rho^B(\omega)$, depends only on the aggregate state and not on how much it lends. The deposit rate is the same for every bank, and the capital requirement depends only on the ratio λ^B. A bank with twice the net worth can therefore double its lending and its deposits, keep the same leverage, and end each period with twice the wealth:

$$W^B(\omega) = n^B \left[\lambda^B \left(\rho^B(\omega) - R \right) + R \right] \tag{27}$$

Its value is then proportional to its net worth in both periods. In period 2, which has no continuation, $V_2^B(n^B, S') = v_2^B(S')\, n^B$, where

$$v_2^B(S') = \max_{\lambda^{B\prime} \leq 1/\theta} \frac{1}{R} \mathbb{E}_{\omega'} \left[\lambda^{B\prime} \left(\rho^{B\prime}(\omega') - R \right) + R \right]. \tag{28}$$

Substituting this into the period-1 problem gives $V_1^B(n^B, S) = v_1^B(S)\, n^B$, and the problem becomes

$$v_1^B(S) = \max_{\lambda^B \leq 1/\theta} \frac{1}{R} \mathbb{E}_{\omega} \left[m^B(\omega) \left(\lambda^B (\rho^B(\omega) - R) + R \right) \right], \tag{29}$$

$$\text{where } m^B(\omega) = \tau_x^B + (1 - \tau_x^B)\, v_2^B\left(S'(\omega)\right), \tag{30}$$

where, $m^B(\omega)$ is the value of one unit of bank wealth in state ω.

The period-1 objective is linear in λ^B, so the bank either lends as much as the requirement allows or lends less. It lends up to the requirement when the value of one more unit of lending is positive:

$$\frac{1}{R} \mathbb{E}_{\omega} \left[m^B(\omega) \left(\rho^B(\omega) - R \right) \right] > 0. \tag{31}$$

Writing the expectation of a product as the product of expectations plus a covari-

ance,

$$\underbrace{\mathbb{E}[m^B]\ \mathbb{E}\left[\rho^B - R\right]}_{\text{lending incentive}} + \underbrace{\text{Cov}\left(m^B(\omega),\, \rho^B(\omega)\right)}_{\text{precautionary motive}} > 0. \tag{32}$$

The first term is the lending incentive. On average, each unit lent earns more than it costs to fund, by $\mathbb{E}[\rho^B] - R = R\tau$, so this term is positive. The second term is the precautionary motive, and it is negative. Loans do worst in the state when the currency depreciates and aggregate productivity is low and many firms default. But this is also when capital is most valuable to a bank: after a crisis, few banks have capital left, so a bank that still has some can lend at high rates. Lending more therefore means losing the most exactly when the bank can least afford to. If this caution outweighs the gain from lending, the bank holds more capital than the requirement asks for. At the calibrated parameters, the gains from lending is greater and so the requirement binds.

4.3.4 The macroprudential wedge

Since every bank lends up to the requirement, aggregate bank net worth at the start of period 2 is

$$N^{B\prime}(\omega) = (1 - \tau_x^B)\ \max\left\{\frac{\rho^B(\omega) - R}{\theta} + R,\ 0\right\} N^B + \xi^B N^B. \tag{33}$$

The maximum reflects limited liability: a bank whose losses exceed its net worth ends the period with nothing.

I start by assuming the economy at a stationary point: absent any shocks, expected bank net worth stays the same from one period to the next.[18] Let $R^e \equiv \mathbb{E}_\omega[W^B(\omega)]/n^B$ denote the expected gross return on bank equity. When the bank survives in every state, stationarity requires

$$(1 - \tau_x^B)\, R^e + \xi^B = 1, \qquad \text{so} \qquad R^e = \frac{1 - \xi^B}{1 - \tau_x^B}.$$

The return on equity is therefore pinned by the entry and exit of bankers. I calibrate R^e to the long run return on equity of Indian banks and ξ^B to equity injections, and τ_x^B drops out as a residual from this condition.

The bank's balance sheet gives a second expression for the same return. With $b_d =$

[18] This is equivalent to the long run or steady state in an infinite period model

n^B/θ, expected wealth per unit of equity is

$$R^e = \frac{1}{\theta}\Big(\mathbb{E}[\rho^B] - R\Big) + R = R\Big(1 + \frac{\tau}{\theta}\Big).$$

Each unit of equity supports $1/\theta$ units of lending, and each unit of lending earns the margin $R\tau$. Equating the two expressions gives the wedge:

$$\tau = \theta\,\frac{R^e - R}{R}. \tag{34}$$

Equity is more expensive than deposits, by the equity premium $(R^e - R)/R$. Requiring banks to fund a larger share of each loan with equity raises the cost of lending, and the wedge rises in proportion to θ. [19]

4.3.5 Solvency floor

A bank survives state ω if its wealth is not negative, $W^B(\omega) \geq 0$. With $n^B = \theta\, b_d$, this is

$$b_d\big(\rho^B(\omega) - R\big) + R\,\theta\, b_d \geq 0 \quad \Longleftrightarrow \quad \theta \geq 1 - \frac{\rho^B(\omega)}{R}.$$

The condition is hardest to meet in the worst realization of ω, where the return on loans is lowest.

Let ω_c denote the worst realization of the state ω. The bank is solvent in every state if and only if $\theta \geq \theta_{\text{floor}}$, where

$$\theta_{\text{floor}} = 1 - \frac{\rho^B(\omega_c)}{R}.$$

Below the floor, the worst realization of ω completely wipes out the bank's equity. Deposit insurance has to then cover the shortfall, $\max\{-W^B(\omega_c), 0\}$, so depositors are repaid in full. I treat this payment as a cost to the government. The regulator never lowers the requirement below this floor.

4.4 The return on capital

Each firm takes the return on capital R^K as given. In the aggregate, it depends on how much capital the economy uses. Capital works with a fixed supply of one unit of labor,

[19] The wedge depends only on the return on equity, not on how stationarity is split between entry and exit.

so aggregate output from capital is $\mathcal{A}K^{\alpha}$, where K is aggregate capital, $\alpha \in (0,1)$ is the capital share and $\mathcal{A} > 0$ is a constant. A unit of capital returns itself plus its marginal product:

$$R^K = 1 + \alpha \mathcal{A} K^{\alpha - 1}. \tag{35}$$

Since $\alpha < 1$, each unit of capital earns less when firms as a whole install more.

Each firm is small, so it takes R^K as given and still operates with constant returns to scale. Only the economy as a whole runs into diminishing returns. Since every firm chooses the same leverage, aggregate capital is $K = \lambda N^F$.

This gives the model a natural recovery force. After a bad shock, firms enter period 2 with less net worth and install less capital, so each unit of capital earns more. Firms coming out of a crisis therefore rebuild faster, and a unit of firm net worth is worth more in that state.

4.5 Timing

Events within the two periods unfold as follows.

Period 1.

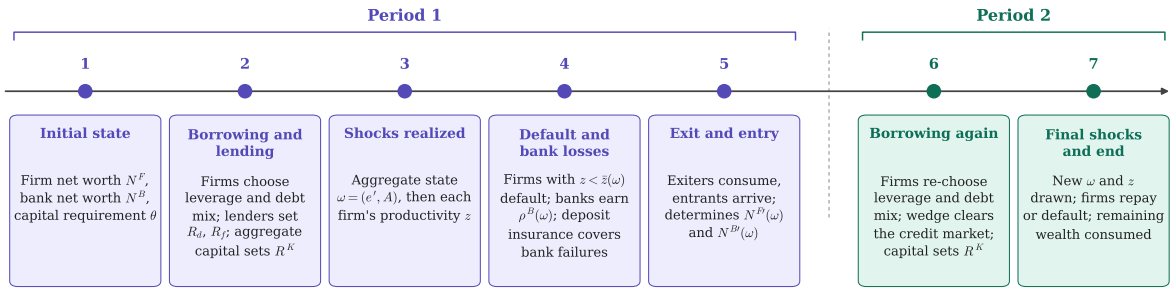
1. The economy starts with aggregate firm net worth N^F and bank net worth N^B, under a capital requirement θ.

2. Firms choose their leverage and debt mix. Lenders set the interest rates R_d and R_f and lend based on the contracted rates.Aggregate capital determines the return on capital R^K.

3. The aggregate state $\omega = (e', A)$ is realized, followed by each firm's own productivity z.

4. Firms whose productivity falls below $\bar{z}(\omega)$ default, and lenders recover part of the project value. Banks earn $\rho^B(\omega)$ on their loans. If a bank's losses exceed its net worth, deposit insurance repays its depositors.

5. Some firms and bankers exit and consume their wealth, and new ones enter. This determines firm and bank net worth at the start of period 2, $N^{F\prime}(\omega)$ and $N^{B\prime}(\omega)$.



Period 2.

6. Firms choose their leverage and debt mix again. The wedge adjusts so that the credit banks can supply equals what firms want to borrow, and aggregate capital determines the return on capital.

7. A new aggregate state and new productivity draws are realized. Firms repay or default, all remaining wealth is consumed, and the economy ends.

The diagram below depicts the timing and flow visually

4.6 What happens after a shock

An adverse shock ω at the end of period 1 damages both banks and firms. At the start of period 2, firms inherit their net worth, $N^{F\prime}(\omega)$ and banks inherit their net worth $N^{B\prime}(\omega)$ following from equations 33 and 17. The wedge τ adjusts to clear the market.

Supply. Banks lend at most what the requirement θ allows, so the credit they can supply is

$$B^{supply}(\omega) = \frac{N^{B\prime}(\omega)}{\theta}.$$

Demand. Each unit of firm net worth borrows $b(\tau')$ from banks, an amount that falls as the wedge rises. Aggregate demand is

$$B^{demand}(\omega) = N^{F\prime}(\omega)\, b\big(\tau'(\omega)\big).$$

A bad shock lowers firm net worth too, so demand for credit falls along with the supply.

Clearing. The wedge $\tau'(\omega)$ adjusts so that demand equals supply. If banks can supply more than firms want even with no wedge, then $\tau'(\omega) = 0$ and banks lend less than the requirement allows. After a mild shock, banks have enough net worth to meet demand at a low wedge or none. The same phenomenon happens if the bank is well capitalized. After a severe shock or if the bank is inadequately capitalized, they fall short, the wedge rises, and firms pay more for domestic credit in period 2.

4.7 Equilibrium

I normalize firm net worth in period 1 to one, $N^F = 1$. Since every choice is proportional to net worth, this is without loss of generality. Households supply deposits elastically at R and foreign lenders supply funds elastically at R^*, so neither market needs a clearing condition.

Definition 1 (Equilibrium)**.** Given a capital requirement θ and a policy regime, an equilibrium consists of

- **allocations:** firm choices (λ, s) and bank leverage (λ^B) in period 1, and, for each aggregate state ω, firm choices (λ', s') and bank leverage $(\lambda^{B\prime})$ in period 2;
- **prices:** interest rates (R_d, R_f), the wedge τ and the return on capital R^K in period 1, and, for each ω, interest rates (R'_d, R'_f), the wedge τ' and the return on capital $R^{K\prime}$ in period 2;
- **aggregate net worth:** bank net worth N^B in period 1, and firm and bank net worth $(N^{F\prime}, N^{B\prime})$ at the start of period 2;
- **values:** the values of firm and bank net worth in each period, ν_1^F, ν_2^F, ν_1^B and ν_2^B,

such that:

1. **Firms** solve the firm's problem in each period, taking prices as given: in period 1 with the weights $m(\omega)$, and in period 2 with $m = 1$.
2. **Banks** solve the bank's problem in each period, taking prices as given, subject to the requirement θ in period 1 and 2.
3. **Interest rates** satisfy the lender's participation condition in equation 10 for each period.

4. **The domestic credit market clears:** the domestic debt firms demand equals the domestic credit banks supply,

$$s(\lambda - 1)\, N^F = \lambda^B N^B \qquad \text{and} \qquad s'(\lambda' - 1)\, N^{F\prime}(\omega) = \lambda^{B\prime}(\omega)\, N^{B\prime}(\omega),$$

with $\tau'(\omega) \geq 0$.

5. **The return on capital** satisfies equation 35, with $K = \lambda N^F$ in period 1 and $K_2 = \lambda' N^{F\prime}$ in period 2.

6. **The wedge** satisfies equation 34

7. **Net worth** evolves according to equations 17 and 33.

8. **Expectations are consistent:** the values $v_2^F(S'(\omega))$ and $v_2^B(S'(\omega))$ that firms and banks use equal the period-2 values in state ω.

In period 1, the wedge is set by equation 34, so the market clearing condition determines aggregate bank net worth. Once that is set, in period 2, aggregate bank net worth is inherited from period 1, so the market clearing condition determines how steep the wedge is.

4.8 Welfare

All agents are risk neutral, so I measure welfare as the sum of the expected values of the domestic agents that policy affects, in period-1 present value:

$$\mathcal{W}(\theta) = \underbrace{v^F N^F}_{\text{firms}} + \underbrace{\tau B_d + \frac{1}{R}\,\mathbb{E}_\omega\big[\tau'(\omega)\, B_d'(\omega)\big]}_{\text{bank owners}} - \underbrace{\frac{1}{R}\,\mathbb{E}_\omega\big[DI(\omega)\big]}_{\text{cost of bank failure}}, \tag{36}$$

where B_d and $B_d'(\omega)$ are domestic credit in periods 1 and 2, and $DI(\omega) = \max\{0, -\big[(\rho^B(\omega) - R)/\theta + R\big]\}N^B$ is the deposit insurance payout.

The first term is the value of firms. It already includes what firms expect in period 2 and the consumption of firms that exit through the weights $m(\omega)$. The second term is bank rents: what banks earn above their cost of deposits. The wedge itself is just a transfer: what firms pay in extra interest, the bank owners receive. However, welfare depends on the wedge through the distortion it creates. A higher wedge pushes firms toward foreign debt, raises their default risk in states where a currency depreciation

occurs, and rations credit after a bad shock. The last term is the cost of bank failure. Below the solvency floor, a crisis state wipes out bank equity, and the deposit insurance becomes active and repays depositors. Covering this shortfall uses public funds, and I count the payout as the cost of bank failure. Above the solvency floor the payout is zero.

Optimal policy. The regulator chooses θ in period 1 to maximize $\mathcal{W}(\theta)$. A tighter requirement has costs: it raises the wedge in period 1, which pushes firms toward foreign debt and raises default after a depreciation. It also has benefits: banks keep more net worth after a bad shock, so less credit is rationed in period 2, and a requirement above the solvency floor rules out bank failure. The optimal requirement balances the two.

5 Calibration

For the calibration, parameters with a direct counterpart in Indian data are set to it. Five parameters without a direct counterpart are chosen so that the model matches the features of Indian firms and banks that drive its mechanism: how much firms borrow, how much of it is in foreign currency, how often they default, how much more they default in bad times, and what they pay for domestic bank credit. Moments that are not targeted serve as a check. The baseline capital requirement is $\theta = 0.09$, the RBI's minimum CRAR. The implied wedge is $\tau = 0.0634 \times 0.09 = 0.57\%$.

Table 3: Parameters set directly

Parameter	Description	Value	Source
R	Domestic real risk-free rate, annual	1.025	Real 10-year G-Sec yield, RBI
R^*	Foreign real rate, annual	1.015	India minus US real rate differential
R^e	Real return on bank equity	1.09	Median return on equity of scheduled commercial banks, 2000 to 2025, RBI
ξ^B	Bank equity inflow	0.041	Equity raised by public sector banks, FY2009 to FY2019, over bank net worth
α	Capital share	0.30	Standard value
σ_e	Exchange rate volatility	0.07	Annual changes in the INR/USD rate, RBI
θ	Capital requirement	0.09	RBI minimum capital ratio
χ	Leverage adjustment cost	2	Limits the response of total borrowing to the wedge
ξ^F	Firm entry	0	See text

Matching the data. The five parameters in Table 4 are solved jointly, but each is tied mainly to one moment. The curvature ψ of the recovery wedge sets the share of debt in domestic currency, the productivity constant $\mathcal{A}$ sets leverage, idiosyncratic risk σ_z

Table 4: Targeted moments

Moment	Target	Source	Parameter	Value
Domestic share of debt, s	0.62	Prowess, firms firms w/ access to foreign credit	Curvature of recovery wedge, ψ	1.879
Leverage, k/n^F	2.40	Prowess, firms w/ access to foreign credit	Productivity constant, $\mathcal{A}$	0.360
Average default rate	4.25%	Slippage ratio, RBI	Idiosyncratic risk, σ_z	0.288
Default rate, bad minus good times	4.5 pp	Slippage ratio, RBI	Aggregate shock, σ_A	0.078
Domestic loan spread, $R_d - R$	1.7 pp	Prowess, large borrowers	Recovery share, κ	0.85

sets the average default rate, the aggregate shock σ_A sets how much default rises in bad times, and the recovery share κ sets the domestic loan spread.

Default rates. I measure default with the slippage ratio, which the RBI defines as the loans that become non-performing during a year as a share of performing loans at the start of the year. Like the default rate in the model, it is a flow. The ratio was 1.4 per cent in 2024–25, and it averaged about 6.5 per cent over 2015–16 to 2017–18, peaking at 7.6 per cent.I target an average default rate of 4.25 per cent and a gap of 4.5 percentage points between bad and good times, which correspond to about 2 per cent in normal years and 6.5 per cent at the peak.

Recovery in default. I choose κ so that the domestic loan spread is 1.7 percentage points, the spread paid by large Indian firms, which are the firms able to borrow abroad. This requires $\kappa = 0.85$, which is higher than recovery rates under India's Insolvency and Bankruptcy Code. The two measure different things. κ is the share of a defaulting firm's project value that lenders recover. Measured recovery rates compare what lenders receive with what they are owed, and a firm defaults precisely because what it owes exceeds what it is worth. A κ as low as measured recovery rates would imply higher spreads than large firms pay and I treat this parameter as one to target the spreads within the range observed in the data.

The aggregate shock. The shock A scales the realized return on capital. The model has no separate price of capital, so A captures both changes in profitability and the revaluation of installed capital, which in models with an explicit price of capital operates through that price. Its volatility is therefore set by the swing in default rates rather than by the volatility of profit rates.

Entry, exit and residuals. The banker exit rate follows from the stationarity condition, $(1 - \tau_x^B)R^e + \xi^B = 1$, which gives $\tau_x^B = 0.120$ as a residual. On the firm side, stationarity, $(1 - \tau_x^F)R^{e,F} + \xi^F = 1$, pins down only one combination of exit and entry.

Firms that enter at the end of period 1 would arrive after the shock, without the currency exposures it acts on. I set $\xi^F = 0$, so that firm net worth in period 2 consists of the balance sheets the shock actually hit, and let the exit rate clear the condition. This gives $\tau_x^F = 0.087$, an expected firm life of about twelve years. Finally, $\bar{\lambda}$ equals leverage at the baseline, 2.40, so the adjustment cost is zero there.

Untargeted moments. Table 5 compares moments that the calibration does not target with the data. At the calibrated parameters, the lending incentive also exceeds the precautionary motive, so banks lend up to the requirement.

Table 5: Untargeted moments

Moment	Model	Data	Source
Real return on capital	5.9%	5.1 to 7.8%	Return on capital employed, 11.4 to 12.6%, less inflation of 4.5 to 6%
Return on capital over the risk-free rate	3.4 pp	2.6 to 5.3 pp	Same, less the real G-Sec yield
Foreign debt spread, $R_f - R^*$	1.6 pp	About 1.7 pp	RBI, external commercial borrowing
Solvency floor, θ_{floor}	0.031	0.03	Leverage ratio: Basel III minimum 3%, RBI 3%

6 Results

This section shows how firms and banks respond to a tighter capital requirement, and how the regulator weighs these responses in choosing the requirement.

6.1 Firms' response to tightening

A higher requirement raises the wedge on domestic credit and firms respond by shifting toward foreign currency debt. Figure 7 shows the domestic share of debt at each wedge. With no wedge, firms borrow 70 per cent of their debt at home. At the baseline wedge of 0.57 per cent this falls to 62 per cent, and at a wedge of 1.5 per cent to 52 per cent.

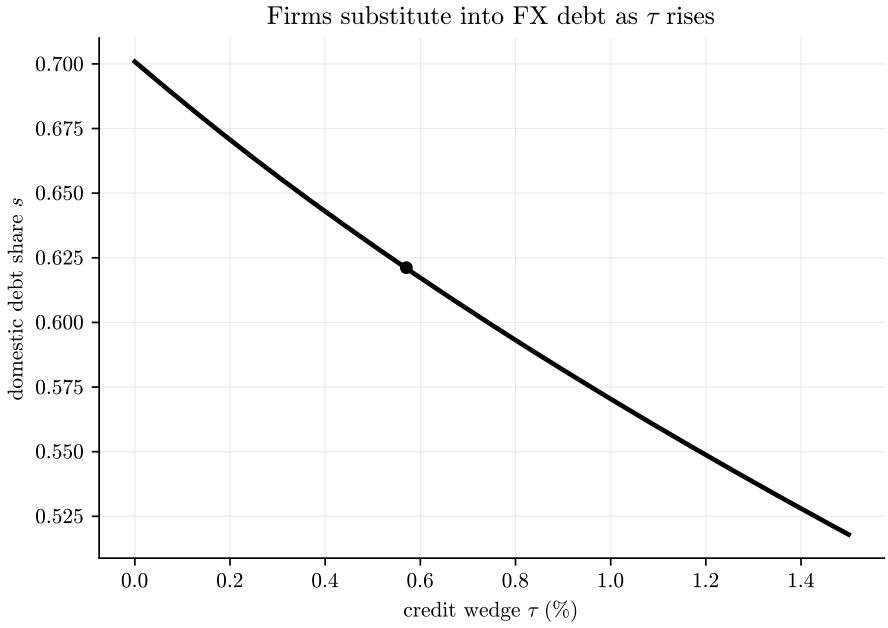


Figure 7: Domestic share of debt against the credit wedge.

Notes: This figure plots the domestic share of debt against the credit wedge in the model. The dot marks the baseline wedge corresponding to a $\theta = 0.09$ that gives $\tau = 0.57$ per cent.

Figure 8 shows default probabilities in a state of currency depreciation as well as appreciation. As the wedge rises from zero to 1.5 per cent. The default probability after a depreciation rises by about 10 per cent, from 4.82 to 5.32 per cent, while the default probability after an appreciation falls by about 8 per cent, from 3.57 to 3.29 per cent. The average rises by about 2.5 per cent, from 4.19 to 4.30 per cent. Tightening therefore moves default risk into the states where the currency depreciates.

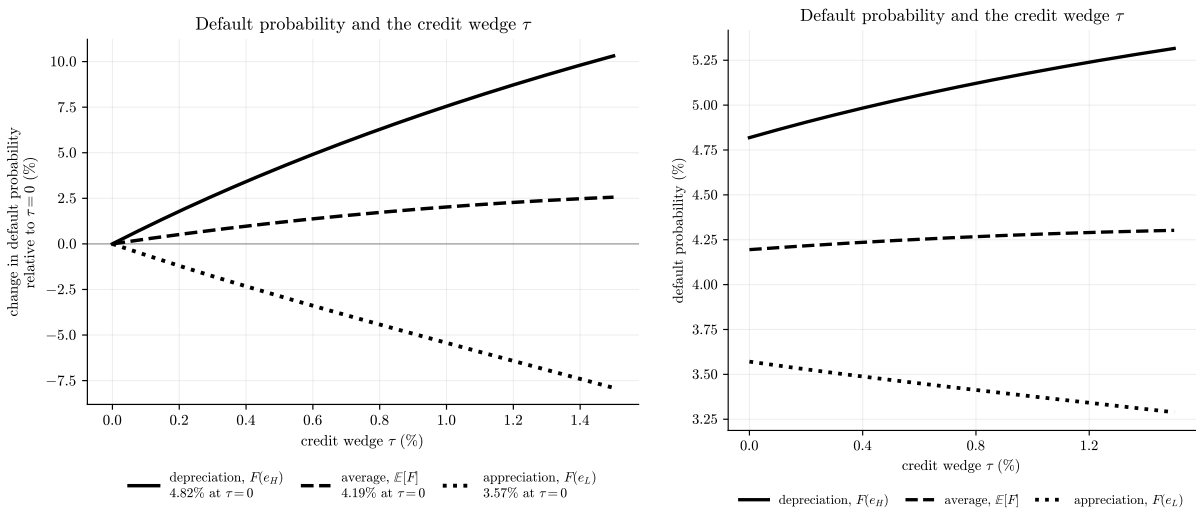


Figure 8: Default probability against the credit wedge.

Notes: This figure plots the default probability of firms against the credit wedge in percent change from $\tau = 0$ (left) and in levels (right).

6.2 Banks and lending after a crisis

The benefit of a higher requirement appears after an adverse shock. Figure 9 shows the credit wedge τ in the state when the currency depreciates and productivity is low (henceforth, referred to as the crisis state). Better capitalized banks ration less after an adverse shock: the crisis wedge falls from 2.8 per cent at $\theta = 0.056$ to 1.4 per cent at the baseline and 0.9 per cent at $\theta = 0.15$. It never reaches zero. A higher requirement gives banks more capital going into the crisis but it also requires more capital behind every loan afterwards.

How much banks lend also depends on the requirement. The requirement caps lending at N^B/θ, but a bank can lend less and hold a voluntary buffer as explained in equation 32. A bank chooses to lend up to the limit when the lending incentive outweighs the precautionary motive. Below a requirement of 5.6 per cent, the precautionary motive wins. Banks hold more capital than what the regulation mandates, and the requirement has no bite. To avoid results driven by the finite horizon of the model, I focus on requirements of 5.6 per cent and above, where banks lend up to the limit.

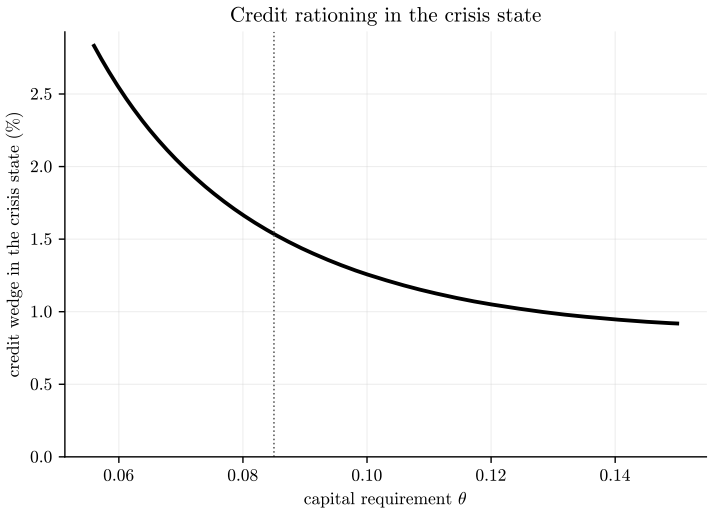


Figure 9: Credit wedge in the crisis state against the capital requirement.

Notes: This figure plots the credit wedge τ against the capital requirement in a crisis state

6.3 Optimal policy

Welfare adds up the value of firms, the rents earned by banks and the cost of deposit insurance. Since every requirement I consider is above the solvency floor of banks at 3.1 per cent, they stay solvent after every shock and the deposit insurance never pays out. Welfare is then the sum of firm value and bank rents.

6.3.1 Bank Rents and Market Failure

Total bank value or the total value of bank rents come from three places. Bank rents in normal times before any shocks hit, and bank rents after good or bad shocks are realized. Figure 10 plots each of them and Figure 11 plots the total bank value.

Normal Times: Before any shock, the wedge is simply the cost of the capital requirement. Banks must fund a share θ of each loan with expensive equity instead of cheap deposits. Each unit of equity earns the same return at every requirement. higher requirement means more equity behind every loan, and firms pay the return on all of it. Rents in normal times therefore rise steadily with θ.

After Bad Shocks: When aggregate productivity is low, more firms default and banks lose part of their capital. Capital is now scarcer and the wedge becomes the price of

that scarcity. It rises until firms borrow only what bank capital can support. If a bank enters the crisis state with more capital, they can ration less and so the wedge is smaller. Therefore, when capital requirements are higher, the wedge and subsequently the rents fall.

After good shocks: When productivity is high, firms repay and banks come out with plenty of capital. Up to a requirement of about 6.5 per cent, banks have more than enough to fund every loan firms want, so there is no wedge and no rent. Above it, the requirement binds even after a boom. Firms with high net worth want to borrow more than banks are allowed to lend, and a wedge appears.

Total: Between 5.6 and 6.5 per cent, the rents banks lose after bad shocks roughly offset the rents they gain in normal times, and total rents are flat. Above 6.5 per cent, rents after good shocks switch on and total rents climb with the requirement. From about 10 per cent on, banks earn more after good shocks than after bad ones.

The dashed line highlights a market failure. A bank keeps capital for a crisis because in this state, capital is worth a lot. It allows the bank to lend at a high crisis wedge but when a bank decides how much capital to keep, it only thinks about its own profit. If all banks kept more capital, the crisis wedge would fall and every firm could borrow more cheaply after a crisis. That gain goes to firms and not to the bank that keeps the capital, so no bank keeps capital for it. Banks as a group don't mind this either, because a shortage of capital after a crisis is what lets them charge a high wedge. On their own, banks never keep more than about 5.6 per cent. The requirement fixes this by making every bank keep more capital, so the crisis wedge falls. The falling dashed line shows this fix at work.

Since banks are competitive, total bank rents rise with the requirement even though no bank wants to keep more on its own. A bank that keeps extra capital by itself cannot charge more for it since firms would borrow from other banks. So extra capital only costs the bank money. When the requirement makes every bank keep more, firms have no cheaper bank to go to, and all banks charge more at once.

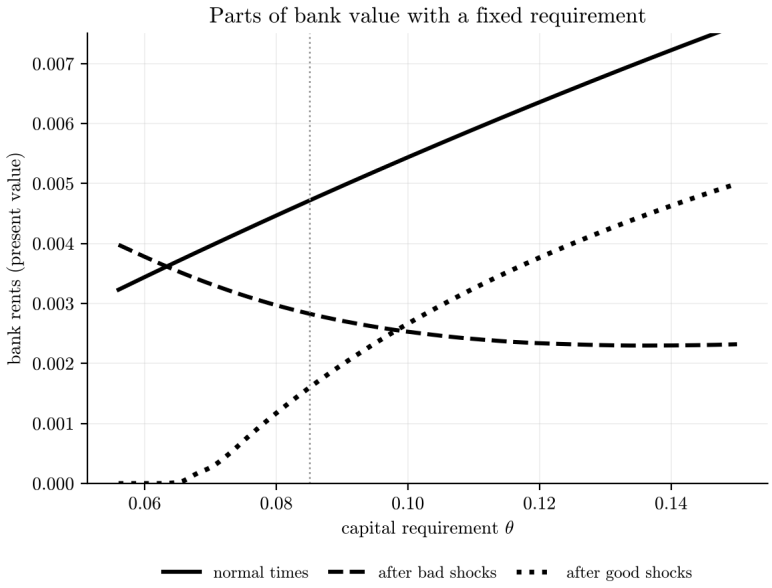


Figure 10: Components of Bank Value

Notes: This figure plots the bank rents earned by the bank for any capital requirement θ in normal times (solid line), after a bad shock (dashed line) and after a good shock (dotted line). Bad shocks are the two states with low productivity, and good shocks the two states with high productivity. The two period-2 lines show the expected present value of rents earned in those states, so the three lines add up to total bank rents.

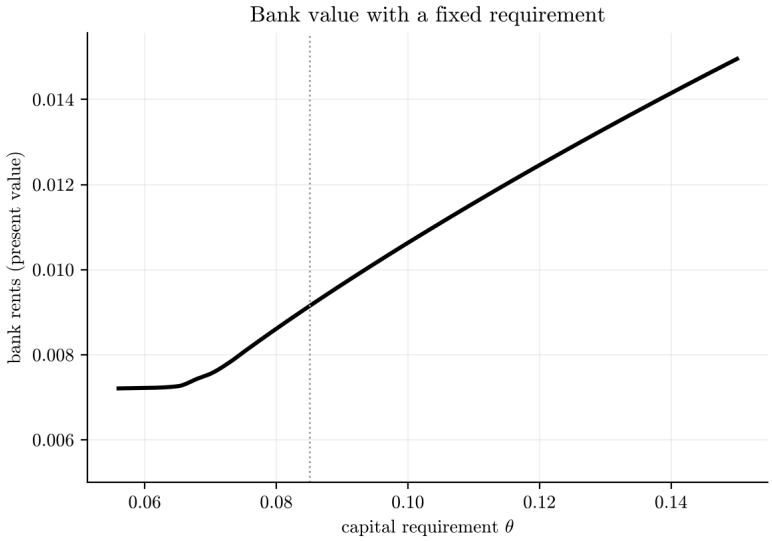


Figure 11: Total Bank Value

Notes: This figure plots the total bank rent earned by the bank for any capital requirement θ.

6.3.2 Firm Value

Figure 12 shows how firm value changes with the requirement θ. Firms are forward looking and gain from a lower crisis wedge. They pay for it in two ways. A higher wedge in normal times pushes them towards foreign currency debt leaving them more exposed to depreciation. And above a requirement of 6.5 per cent, firms also pay a wedge after good shocks. Firm value rises up to 6.5 per cent after which it falls monotonically.

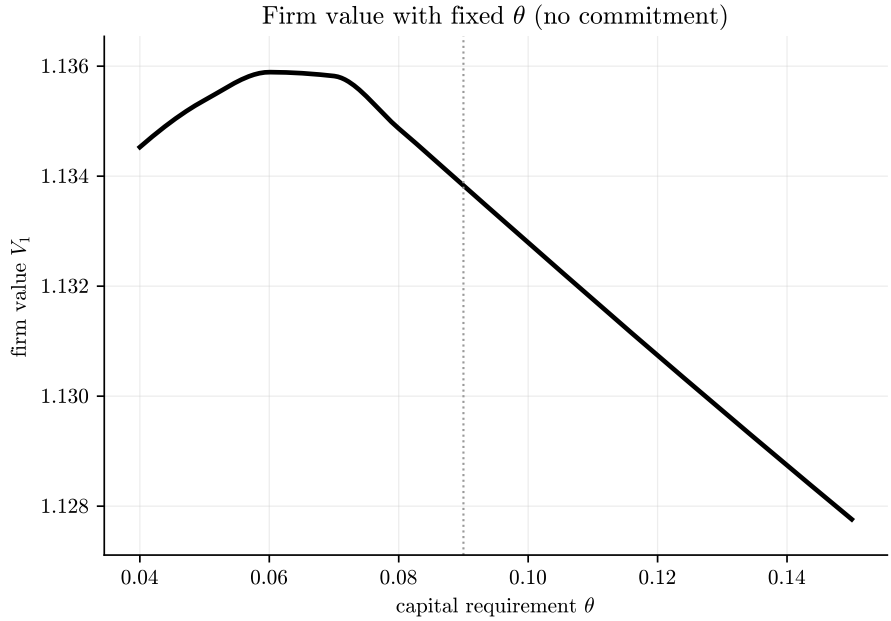


Figure 12: Firm Value

Notes: This figure plots the firm value for any capital requirement θ.

6.3.3 Optimal Requirement

Figure 13 adds the two. Welfare peaks at $\theta = 0.084$. Between 6.5 per cent and the optimum at 8.4 per cent, firm value falls less than what bank rents rise by. This difference is a real gain. With better capitalized banks, firms keep investing after a crisis. Above the optimum, firms enter the crisis with more foreign currency debt and less net worth, so investment falls while the cost of the wedge in normal times and after good shocks keep rising. The optimum requirement is in line with RBI's long run regulatory minimum at 9 per cent.

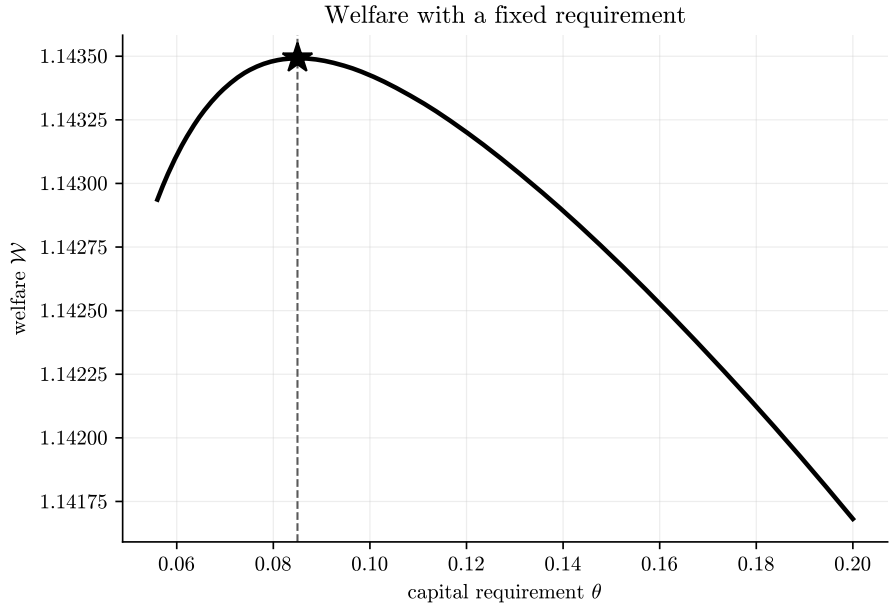


Figure 13: Welfare

Notes: This figure plots the Welfare against the capital requirement, θ. The star and dashed vertical line marks the optimum level at 0.084

Taken together, the results describe a trade-off with an intuitive resolution. A tighter capital requirement raises the cost of domestic credit, pushes firms toward foreign currency debt, and concentrates their default risk in the states where the currency depreciates. It also leaves banks better capitalized going into a crisis, so they keep lending when firms need it most. Banks do not hold this capital on their own, because no single bank gains from the lower borrowing costs its capital brings to every firm. The requirement corrects this market failure. Holding the requirement fixed across good times and bad, the regulator balances these forces at 8.5 per cent, close to the RBI's minimum capital ratio of 9 per cent, which is also fixed.

7 Conclusion

This paper highlights a tension inherent to macroprudential policy: it acts on banks, but its effects do not stop with banks. Measures that strengthen and stabilize the financial system also restrain the credit that firms rely on. Part of the credit they restrain reappears as foreign currency debt on firm balance sheets, and this debt is costly when the currency depreciates. The leakage matters for policy because banks remain ex-

posed to these firms through their balance sheets. By making firms riskier and more prone to default, the policy feeds back onto the banks themselves. Optimal policy therefore requires a careful evaluation of both the firm side and the bank side.

The evidence comes in two steps. Firms that depend more on bank credit respond to a tightening by moving out of bank loans and into foreign currency debt, while their total borrowing stays roughly unchanged. Firms that carry more foreign currency debt then report larger forex losses, lower profits and a higher chance of distress in years when the rupee falls. Taken together, the two steps show that a tightening leaves firms more exposed to a depreciation. Earlier work has traced macroprudential leakage mostly through lenders. These results add a margin on the borrower side, where the leakage takes the form of currency exposure.

I then develop a two-period small open economy model that brings firms and banks together in one framework. The model makes the channels of the trade-off explicit. A higher capital requirement makes banks safer. It also raises the cost of domestic credit, moving firms toward foreign currency debt and increasing their likelihood of default in states where the currency depreciates. Since banks hold the domestic debt of these same firms, the defaults that follow a depreciation impose losses on banks. The requirement still has value because it shifts lending capacity across states. Banks hold more capital and firms pay more for credit in normal times. In return, the wedge on domestic credit is smaller after a crisis, when firms need credit most. Each bank takes the price of credit as given, so it does not count that its capital lowers borrowing costs for every firm after a crisis. Left alone, banks hold too little capital, and the requirement corrects this market failure. Setting the optimal stance of policy therefore requires balancing these two forces. The model calibrated to India suggests that the central bank's minimum capital requirement is close to the optimal level.

The broader lesson is that macroprudential policy cannot be judged by the balance sheets of the regulated banks alone. When firms can borrow elsewhere, tightening domestic credit can move risk rather than remove it. In an open economy, the risk it moves can take the form of currency exposure that can come back to the banking system in a depreciation episode. Hence, for emerging markets where firms have access to foreign currency debt, macroprudential policy needs to account for both the resilience of banks and the currency composition of corporate balance sheets.

The results also point to the importance of accessible and liquid currency hedging markets. The RBI's concern for the stability of the domestic banking system will remain, and so will its use of macroprudential tools. Restricting foreign borrowing

through tighter capital controls does not solve the problem either since firms that lose bank credit need to still invest in productive opportunities. Non-bank lenders can only serve part of the firms' demand for credit and India's corporate bond market is open mainly to highly rated firms. Foreign borrowing is therefore a useful source of funding for firms. The risk lies in leaving it unhedged. Deeper and more liquid hedging markets would let firms that turn abroad protect themselves against a depreciation at a market price, choosing how much to hedge based on their own costs and revenues, so that a tighter capital requirement makes banks safer without leaving the firms they lend to more fragile.

The findings suggest several directions for further work, starting with an endogenous exchange rate. If firm defaults and consequent capital outflows feed back into the exchange rate, the fragility channel would be stronger and more closely tied to the external conditions under which emerging markets borrow. Another extension would be to study seniority between lenders. Domestic and foreign lenders may differ in their seniority over firm assets, which could lead to richer default and restructuring outcomes.

References

Viral V Acharya and Siddharth Vij. Foreign currency borrowing of corporations as carry trades: Evidence from india. Technical report, National Bureau of Economic Research, 2020.

Tobias Adrian, Paolo Colla, and Hyun Song Shin. Which financial frictions? parsing the evidence from the financial crisis of 2007 to 2009. *NBER macroeconomics annual*, 27(1):159–214, 2013.

Toni Ahnert, Kristin Forbes, Christian Friedrich, and Dennis Reinhardt. Macroprudential fx regulations: shifting the snowbanks of fx vulnerability? *Journal of Financial Economics*, 140(1):145–174, 2021.

Shekhar Aiyar, Charles W Calomiris, and Tomasz Wieladek. Does macro-prudential regulation leak? evidence from a uk policy experiment. *Journal of Money, Credit and Banking*, 46(s1):181–214, 2014.

Zohair Alam, Adrian Alter, Jesse Eiseman, Gaston Gelos, Heedon Kang, Machiko Narita, Erlend Nier, and Naixi Wang. Digging deeper—evidence on the effects of

macroprudential policies from a new database. *Journal of Money, Credit and Banking*, 2019.

Laura Alfaro, Gonzalo Asis, Anusha Chari, and Ugo Panizza. Corporate debt, firm size and financial fragility in emerging markets. *Journal of International Economics*, 118:1–19, 2019.

Stefan Avdjiev, Catherine Casanova, Patrick McGuire, and Goetz Von Peter. International prudential policy spillovers: a global perspective. 2016.

Bo Becker and Victoria Ivashina. Cyclicality of credit supply: Firm level evidence. *Journal of Monetary Economics*, 62:76–93, 2014.

Ben S Bernanke, Mark Gertler, and Simon Gilchrist. The financial accelerator in a quantitative business cycle framework. *Handbook of macroeconomics*, 1:1341–1393, 1999.

Marianne Bertrand, Esther Duflo, and Sendhil Mullainathan. How much should we trust differences-in-differences estimates? *The Quarterly journal of economics*, 119(1): 249–275, 2004.

Markus Brunnermeier, Andrew Crockett, Charles AE Goodhart, Avinash Persaud, Hyun Song Shin, et al. *The fundamental principles of financial regulation*, volume 11. ICMB, Internat. Center for Monetary and Banking Studies Geneva, 2009.

Valentina Bruno and Hyun Song Shin. Capital flows and the risk-taking channel of monetary policy. *Journal of monetary economics*, 71:119–132, 2015.

Valentina Bruno and Hyun Song Shin. Global dollar credit and carry trades: a firm-level analysis. *The Review of Financial Studies*, 30(3):703–749, 2017.

Valentina Bruno and Hyun Song Shin. Currency depreciation and emerging market corporate distress. *Management Science*, 66(5):1935–1961, 2020.

Eugenio Cerutti, Stijn Claessens, and Luc Laeven. The use and effectiveness of macroprudential policies: New evidence. *Journal of financial stability*, 28:203–224, 2017.

Michael KF Chui, Ingo Fender, and Vladyslav Sushko. Risks related to eme corporate balance sheets: the role of leverage and currency mismatch. *BIS Quarterly Review September*, 2014.

Janko Cizel, Jon Frost, Aerdt Houben, and Peter Wierts. Effective macroprudential policy: cross-sector substitution from price and quantity measures. *Journal of Money, Credit and Banking*, 51(5):1209–1235, 2019.

Nicolas Crouzet. Aggregate implications of corporate debt choices. *The Review of Economic Studies*, 85(3):1635–1682, 2018.

Wenxin Du and Jesse Schreger. Sovereign risk, currency risk, and corporate balance sheets. *The Review of Financial Studies*, 35(10):4587–4629, 2022.

Barry Eichengreen, Ricardo Hausmann, and Ugo Panizza. The pain of original sin. *Other People's Money: Debt Denomination and Financial Instability in Emerging Market Economies, University of Chicago Press, Chicago*, 2003.

Barry Eichengreen, Ricardo Hausmann, and Ugo Panizza. Currency mismatches, debt intolerance, and the original sin: Why they are not the same and why it matters. In *Capital controls and capital flows in emerging economies: Policies, practices, and consequences*, pages 121–170. University of Chicago Press, 2007.

Barry Eichengreen, Ricardo Hausmann, and Ugo Panizza. Yet it endures: The persistence of original sin. *Open Economies Review*, 34(1):1–42, 2023.

Vadim Elenev, Tim Landvoigt, and Stijn Van Nieuwerburgh. A macroeconomic model with financially constrained producers and intermediaries. *Econometrica*, 89(3):1361–1418, 2021.

Luigi Falasconi, Pablo Herrero, Caterina Mendicino, and Dominik Supera. The foreign liability channel of bank capital requirements. *Available at SSRN 4999829*, 2024.

Gabriele Galati and Richhild Moessner. Macroprudential policy, a literature review. *Journal of Economic Surveys*, 27(5):846–878, 2013. doi: https://doi.org/10.1111/j.1467-6419.2012.00729.x. URL https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-6419.2012.00729.x.

Rohit Goel, Swapnanil SenGupta, and Udaibir Saran Das. Corporate foreign exchange risk in india: firm-level evidence and some policy implications. *Margin: the journal of applied economic research*, 19(2):131–186, 2025.

Samuel G Hanson, Anil K Kashyap, and Jeremy C Stein. A macroprudential approach to financial regulation. *Journal of economic Perspectives*, 25(1):3–28, 2011.

Ricardo Hausmann and Ugo Panizza. On the determinants of original sin: an empirical investigation. *Journal of international Money and Finance*, 22(7):957–990, 2003.

Òscar Jordà. Local projections for applied economics. *Annual Review of Economics*, 15 (1):607–631, 2023.

Yun Jung Kim, Linda L Tesar, and Jing Zhang. The impact of foreign liabilities on small firms: Firm-level evidence from the korean crisis. *Journal of International Economics*, 97(2):209–230, 2015.

Enrique G Mendoza. Macroprudential policy: Promise and challenges. Technical report, National Bureau of Economic Research, 2016.

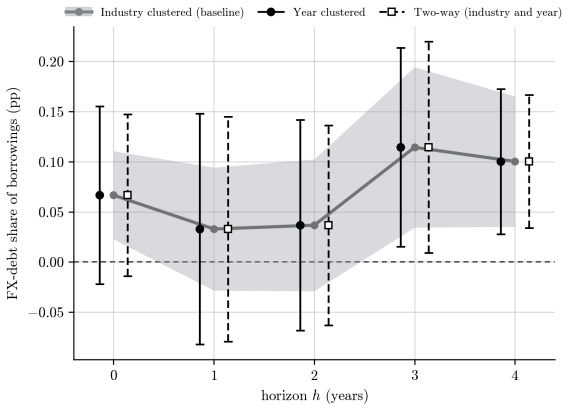
Friederike Niepmann and Tim Schmidt-Eisenlohr. Foreign currency loans and credit risk: Evidence from u.s. banks. *Journal of International Economics*, 135:103558, 2022. ISSN 0022-1996. doi: https://doi.org/10.1016/j.jinteco.2021.103558. URL https://www.sciencedirect.com/science/article/pii/S0022199621001380.

Ila Patnaik, Ajay Shah, and Nirvikar Singh. Foreign currency borrowing by indian firms: Towards a new policy framework, 2015.

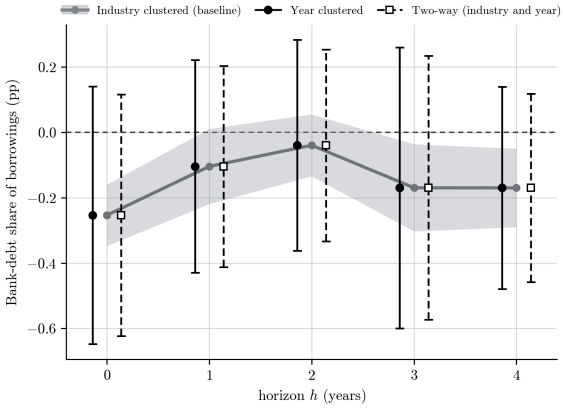
Dennis Reinhardt and Rhiannon Sowerbutts. Regulatory arbitrage in action: evidence from banking flows and macroprudential policy. 2015.

Jin Young Yang and Hyunduk Suh. Heterogeneous effects of macroprudential policies on firm leverage and value. *International Review of Financial Analysis*, 86:102554, 2023. ISSN 1057-5219. doi: https://doi.org/10.1016/j.irfa.2023.102554. URL https://www.sciencedirect.com/science/article/pii/S1057521923000704.

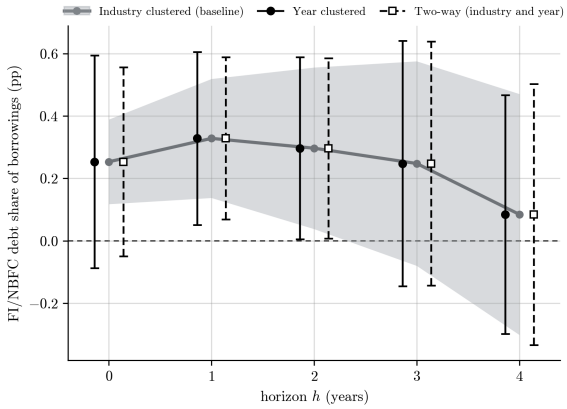
Appendix A



(a) Foreign Currency Debt Share



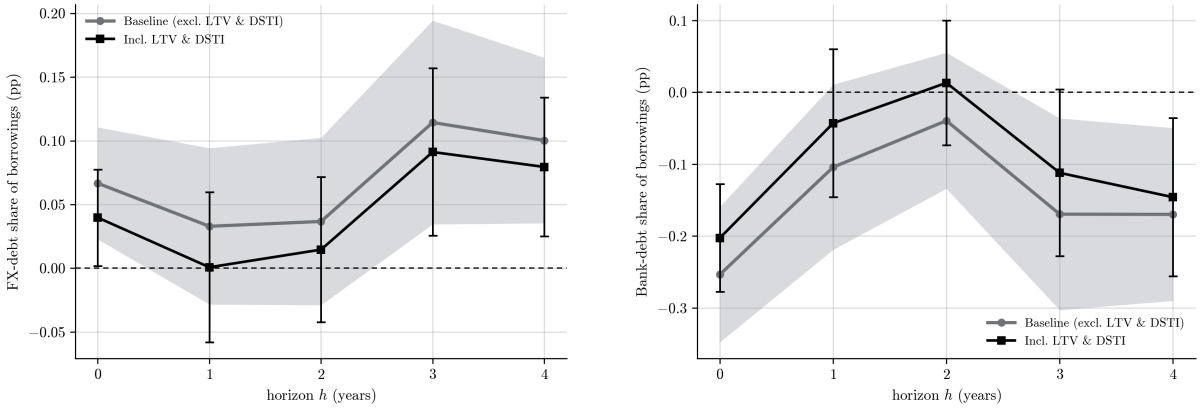
(b) Bank Debt Share



(c) Financial Institutions Share

Figure A.1: Response of Debt Composition Under Different Standard Errors

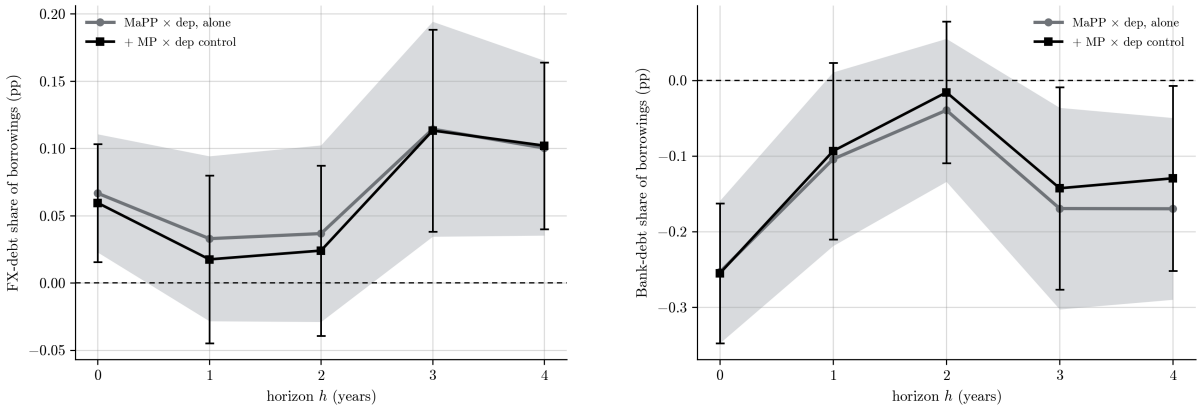
Notes: The figures plot impulse responses of the foreign-currency debt share (a) bank debt share (b) and NBFC share (c) following a macroprudential tightening shock. All regressions include firm and industry by year fixed effects, and standard errors are clustered by industry (shaded grey), year (solid line) and two way firm and time (dotted lines) at 95% confidence intervals.



(a) Foreign Currency Debt Share (b) Bank Debt Share

Figure A.2: Response of Debt Composition to Macroprudential Tightening (including borrower-based instruments)

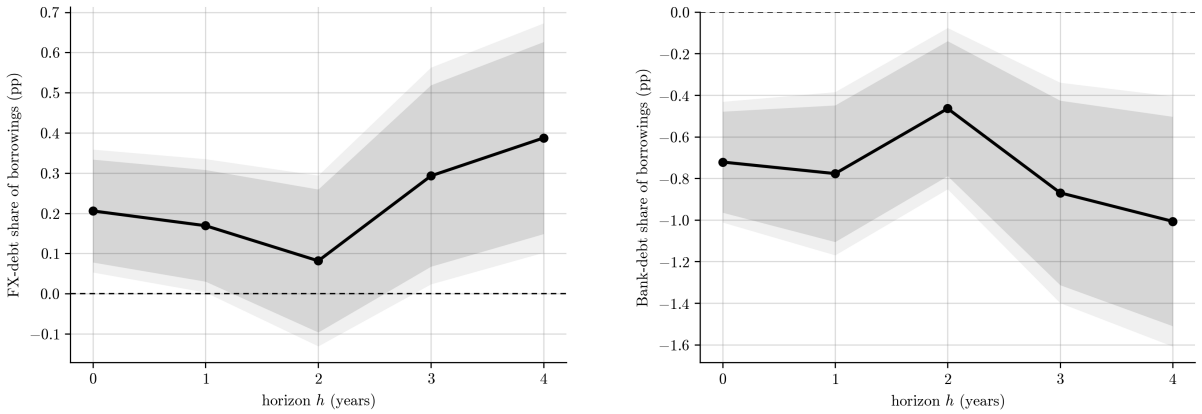
Notes: The figures plot impulse responses of the foreign-currency debt share (a) and bank debt share (b) following a macroprudential tightening shock. The policy instrument includes borrower-based instruments in the black and the baseline in grey. All regressions include firm and industry by year fixed effects, and standard errors are clustered by industry.



(a) Foreign Currency Debt Share (b) Bank Debt Share

Figure A.3: Response of Debt Composition to Macroprudential Tightening controlling for differential effect of domestic monetary policy on bank-dependent firms

Notes: The figures plot impulse responses of the foreign-currency debt share (a) and bank debt share (b) following a macroprudential tightening shock while controlling for the differential impact of domestic monetary policy measured by the repo rate (Source: Federal Reserve Economic Data) on bank dependent firms in black and the baseline in grey. All regressions include firm and year fixed effects, and standard errors are clustered by firm. All regressions include firm and industry by year fixed effects, and standard errors are clustered by industry.



(a) Foreign Currency Debt Share

(b) Bank Debt Share

Figure A.4: Response of Debt Composition to Macroprudential Tightening (as a continuous measure)

Notes: The figures plot impulse responses of the foreign-currency debt share (a) and bank debt share (b) following a macroprudential tightening shock. To capture the intensity of the shock, here, the shock is the standardized change in the cash reserve ratio and the capital conservation buffer over each firm's fiscal year. All regressions include firm and 3-digit industry by year fixed effects, and standard errors are clustered by industry. Darker shaded regions denote 90% and lighter shaded regions 95% confidence intervals.

Appendix B

B.1 Firm Default Threshold Homogeneity

The threshold depends on the firm's balance sheet only through two ratios. Let $\lambda \equiv k/n^F$ denote leverage. Using the balance sheet and the definition of s (with $e = 1$), domestic debt is $b_d = s(k - n^F)$ and foreign debt is $b_f = (1 - s)(k - n^F)$. Substituting gives

$$\bar{z}(\omega) = \frac{\lambda - 1}{\lambda} \cdot \frac{R_d\, s + R_f\, e'(1 - s)}{AR^K}.$$

Net worth cancels, so the threshold depends on leverage and on the mix of debt, but not on the size of the firm.

$$y_{i,t+h} = \alpha_i^h + \delta_{jt}^h + \beta_h \left(\mathrm{CC}_t \times D_i^{2013}\right) + \Gamma_h' \left(\mathrm{CC}_t \times X_i^{2013}\right) + \rho_h\, y_{i,t-1} + \Theta_h' Z_{i,t-1} + \varepsilon_{i,t+h}, \qquad h \in \{-3, -2$$